

Intangible capital and the investment- q relation[☆]Ryan H. Peters^{*}, Lucian A. Taylor

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ABSTRACT

The neoclassical theory of investment has mainly been tested with physical investment, but we show that it also helps explain intangible investment. At the firm level, Tobin's q explains physical and intangible investment roughly equally well, and it explains total investment even better. Compared with physical capital, intangible capital adjusts more slowly to changes in investment opportunities. The classic q theory performs better in firms and years with more intangible capital: Total and even physical investment are better explained by Tobin's q and are less sensitive to cash flow. At the macro level, Tobin's q explains intangible investment many times better than physical investment. We propose a simple, new Tobin's q proxy that accounts for intangible capital, and we show that it is a superior proxy for both physical and intangible investment opportunities.

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1. Introduction

The neoclassical theory of investment was developed more than 30 years ago, when firms mainly owned physical assets such as property, plant, and equipment (PP&E). As a result, empirical tests of the theory have focused almost exclusively on physical capital. Since then, the US economy has shifted toward service- and technology-based industries, which has made intangible assets such as human capital, innovative products, brands, patents, software, customer relationships, databases, and distribution systems increasingly important. Corrado and Hulten (2010) estimate that intangible capital makes up 34% of firms' total capital in recent years. Despite the importance of intangible capital, researchers have almost always excluded it when testing investment theories.

Is there a role for intangible capital in the neoclassical theory of investment? If so, how must empirical tests be adapted? Is the theory still relevant in an economy increasingly dominated by intangible capital? For example, the Hayashi (1982) classic q -theory of investment predicts that Tobin's q , the ratio of capital's market value to its

replacement cost, perfectly summarizes a firm's investment opportunities. As a result, Tobin's q has become "arguably the most common regressor in corporate finance" (Erickson and Whited, 2012, p. 1286). How should researchers proxy for investment opportunities in an increasingly intangible economy? And how well do those proxies work?

To answer these questions, we revisit the basic empirical facts about the relation between corporate investment, Tobin's q , and free cash flow. A very large investment literature, both in corporate finance and macroeconomics, is built upon these fundamental facts, so it is important to understand how the facts change when accounting for intangible capital. We show that some facts do change significantly, and we discuss the implications for our theories of investment. Most important, we show that the classic q theory of investment, despite originally being designed to explain physical investment, also helps explain intangible investment. In other words, the neoclassical theory of investment is still relevant. An important component of our analysis is a new Tobin's q proxy that accounts for intangible capital. We show that this new proxy captures firms' investment opportunities better than other popular proxies, thus offering a simple way to improve corporate finance regressions without additional econometrics.

To guide our empirical work, we begin with a theory of a firm that invests optimally in physical and intangible capital over time. The theory is a standard neoclassical investment- q theory in the spirit of Hayashi (1982) and Abel and Eberly (1994). Like physical capital, intangible capital is costly to obtain and helps produce future profits, albeit with some risk. For this fundamental reason, it makes sense to treat intangible capital as capital in the neoclassical framework. Our theory predicts that a firm's physical and intangible investment rates should both be explained well by a version of Tobin's q that we call "total q ," which equals the firm's market value divided by the sum of its physical and intangible capital stocks.

We test this and other predictions using data on public US firms from 1975 to 2011. We measure a firm's intangible capital as the sum of its knowledge capital and organization capital. We interpret research and development (R&D) spending as an investment in knowledge capital, and we apply the perpetual-inventory method to a firm's past R&D to measure the replacement cost of its knowledge capital. We similarly interpret a fraction of past selling, general, and administrative (SG&A) spending as an investment in organization capital, which includes human capital, brand, customer relationships, and distribution systems. Our measure of intangible capital builds on the measures of Corrado and Hulten (2010, 2014), Corrado, Hulten, and Sichel (2009), Eisefeldt and Papanikolaou (2013, 2014), Falato, Kadyrzhanova, and Sim (2013), Lev and Radhakrishnan (2005), and Zhang (2014). We define a firm's total capital as the sum of its physical and intangible capital, both measured at replacement cost. Guided by our theory, we measure total q as the firm's market value divided by its total capital, and we scale the physical and intangible investment rates by total capital.

While our intangible-capital measure has limitations, we believe, and the data confirm, that an imperfect proxy

is better than setting intangible capital to zero. A benefit of the measure is that it is easily computed for all public US firms back to 1975, and it requires only Compustat data and other easily downloaded data. Our data on firms' total q and intangible capital can be downloaded from Wharton Research Data Services (WRDS).

Our analysis begins with ordinary least squares (OLS) panel regressions of investment on q . Consistent with our theory, total q explains physical and intangible investment roughly equally well. Their within-firm R^2 values are 21% and 28%, respectively. Total q explains the sum of physical and intangible investment (total investment) even better, delivering an R^2 of 33%. Judging by R^2 , the neoclassical theory of investment works at least as well for intangible capital as for physical capital, and it works even better for an all-inclusive measure of capital. Also consistent with our theory, the literature's standard investment regression, which excludes intangible capital, typically delivers lower R^2 values.

According to the theory, physical and intangible investment should co-move, because they share the same marginal productivity of capital, as proxied by total q . The data support this view: The within-firm correlation between physical and intangible investment is 31% but drops to 17% after controlling for total q .

Throughout the corporate finance literature, researchers use Tobin's q to proxy for firms' investment opportunities. Our OLS R^2 values help evaluate these proxies. We find that including intangible capital in our q measure produces a superior proxy for investment opportunities, no matter how we measure investment. We compare total q with the investment literature's standard q measure, which scales firm value by physical capital (PP&E) alone. Total q is better at explaining physical, intangible, and total investment, as well as R&D investment and the literature's standard investment measure, capital expenditure (CAPX) scaled by PP&E. It is also popular to measure Tobin's q as the firm's market value scaled by the book value of assets. The problem with this measure is that "Assets" on the balance sheet excludes the vast majority of firms' intangible capital, because US accounting rules treat R&D and SG&A as operating expenses, not capital investments. Like Erickson and Whited (2006, 2012), we find that market-to-book-assets ratios are especially poor proxies for investment opportunities.

The OLS regressions suffer from two well-known problems. First, the slopes on q are biased due to measurement error in q . Second, the OLS R^2 depends not just on how well q explains investment, but also on how well our q proxies explain the true, unobservable q . To address these problems, we reestimate the investment models using the Erickson, Jiang, and Whited (2014) cumulant estimator. This estimator produces unbiased slopes and a statistic τ^2 that measures how close our q proxy is to the true, unobservable q . Specifically, τ^2 is the R^2 from a hypothetical regression of our q proxy on the true q . We find that τ^2 is 21% higher when we include intangible capital in the investment- q regression, implying that our new q proxy is closer to the true q .

According to our theory, slope coefficients of investment on total q help measure capital adjustment costs.

The inverse q -slope for physical (intangible) investment measures the convex component of physical (intangible) capital's adjustment costs. We find that intangible investment's q -slope is roughly half as large as physical investment's, implying intangible capital's convex adjustment costs are twice as large as those for physical capital. This finding supports the literature's conjecture that intangible capital is costlier than physical capital to adjust, because adjusting intangible capital often requires replacing highly trained employees (e.g., [Brown, Fazzari, and Petersen, 2009](#); [Grabowski, 1968](#)). An important implication of our result is that firms adjust more slowly to changes in investment opportunities as the economy shifts toward intangible capital. We also find that accounting for intangibles roughly doubles the q -slope for physical investment, implying significantly lower convex adjustment costs for physical capital than previously believed.

Like other simple q theories, ours predicts that cash flow should not help explain investment after controlling for q . Researchers typically measure cash flow as profits net of R&D and SG&A. Because R&D and at least part of SG&A are actually investments, one should add them back to measure cash flow available for investment. After making this adjustment, we find that physical investment becomes more sensitive to cash flow than previously believed. On this dimension, the neoclassical theory fits the data worse after accounting for intangibles. In contrast, the R&D component of intangible investment is insensitive to cash flow, supporting the theory. Because SG&A's investment component is difficult to measure, it remains unclear whether intangible investment overall is more sensitive than physical investment to cash flow. Financing constraints are unlikely to explain the opposing cash flow results for physical and R&D capital, as financing constraints are arguably more severe for R&D capital due to its lower collateral value ([Almeida and Campello, 2007](#); [Falato, Kadyrzhanova, and Sim, 2013](#)). More recent theories predict an investment–cash flow sensitivity even without financing constraints.¹ For example, diseconomies of scale can make cash flow informative about investment opportunities, even controlling for Tobin's q . Without a full structural estimation, it is difficult to tell whether our cash flow results are driven by differences in financing constraints, diseconomies of scale, or some other source.

Several important investment studies use data only from manufacturing firms.² Surprisingly, we find that the classic q theory fits the data better outside the manufacturing industry and, more generally, in firms and years with more intangible capital. Investment is usually better explained by q and is less sensitive to cash flow in subsamples with more intangibles. These results even hold using the literature's standard measures that exclude intan-

gibles. Again, our results imply that the neoclassical theory of investment is just as relevant, if not more so, in an increasingly intangible economy. Why the theory fits better in high-intangible settings remains unclear. We find no robust evidence that high-intangible firms are closer to the theory's ideal of perfect competition and constant returns to scale. Also, high-intangible firms arguably face more financing constraints, which should make theory fit worse, not better.

Some of our main results are even stronger in macroeconomic time series data. For example, the literature's standard investment– q regression, which excludes intangibles, delivers an R^2 of just 4%, whereas the regression including intangible capital produces an R^2 of 61%. In first differences, total q explains physical and intangible capital roughly equally well. Again, the neoclassical theory of investment applies just as well, if not better, to intangible capital.

The empirical investment– q literature is extensive and dates back at least to [Ciccolo \(1975\)](#) and [Abel \(1980\)](#). [Hassett and Hubbard \(1997\)](#) and [Caballero \(1999\)](#) review the literature. Tests of the classic q theory using physical capital have been disappointing. Investment is typically sensitive to cash flow, is explained poorly by q (low R^2), and produces implausibly large adjustment-cost parameters (low q -slopes). We show that including intangible capital helps solve the last two problems but not the first one. Other attempts to solve these problems with better measurement include using a fundamental q instead of market values directly ([Abel and Blanchard, 1986](#)), using bond prices ([Philippon, 2009](#)), correcting for measurement error ([Erickson, Jiang, and Whited, 2014](#); [Erickson and Whited, 2000, 2012](#)), and using state variables directly ([Gala and Gomes, 2013](#)). We also correct for measurement error, and we show that including intangibles yields even larger improvements than using bond prices.

Our paper is not the first to examine the empirical relation between intangible investment and Tobin's q . [Eisfeldt and Papanikolaou \(2013\)](#) find a positive relation between investment in organization capital and q . [Almeida and Campello \(2007\)](#) and others use q and cash flow to forecast R&D investment. [Chen, Goldstein, and Jiang \(2007\)](#) use q to forecast the sum of physical investment and R&D. Closer to our specifications, [Baker, Stein, and Wurgler \(2002\)](#) measure investment as the sum of CAPX, R&D, and SG&A, and they regress them on q . [Gourio and Rudanko \(2014\)](#) examine the relation between q and investment in customer capital, a type of intangible capital. All these papers use a q proxy that excludes intangibles from the denominator. Besides having a different focus, our paper is the first to include all types of intangible capital not just in investment, but also in Tobin's q and cash flow. Including intangibles in all three measures is important for delivering our results. [Belo, Lin, and Vitorino \(2014\)](#) show that physical and brand investment are both procyclical, which is related to our comovement result, but they do not examine Tobin's q .

[Almeida and Campello \(2007\)](#) examine how asset tangibility and financial constraints affect the investment–cash flow relation. Like us, they find a higher investment–cash flow sensitivity for firms using less intangibles. Unlike our measures of asset intangibility, theirs exclude firms'

¹ Examples include [Gomes \(2001\)](#), [Alti \(2003\)](#), [Cooper and Ejarque \(2003\)](#), [Hennessy and Whited \(2007\)](#), [Abel and Eberly \(2011\)](#), [Gourio and Rudanko \(2014\)](#), and [Abel \(2016\)](#).

² Examples include [Fazzari, Hubbard, and Petersen \(1988\)](#), [Almeida and Campello \(2007\)](#), and [Erickson and Whited \(2012\)](#). A common reason is that manufacturing firms' capital is easier to measure. Our τ^2 statistics confirm that the literature's standard q proxy has less measurement error in the manufacturing industry compared with other industries.

internally created intangible assets, which we find make up the vast majority of intangible capital.

Li, Liu, and Xue (2014) structurally estimate a q -theory model that includes intangible capital. Like us, they find that intangible capital has larger adjustment-cost parameters than physical capital and that including intangibles decreases physical capital's estimated adjustment costs. Unlike us, they focus on the cross section of stock returns, and they exclude organization capital.

The paper proceeds as follows. Section 2 presents our theory of investment in physical and intangible capital. Section 3 describes the data and intangible-capital measure we use to test the theory's predictions. Section 4 presents full-sample results, and Section 5 compares results across different types of firms, industries, and years. Section 6 contains results for the overall macroeconomy. Section 7 explores the robustness of our empirical results, and Section 8 concludes.

2. Intangible capital and the neoclassical theory of investment

In this section, we review the neoclassical theory of investment, and we argue that intangible capital fits well into the theory. We simplify and modify the Abel and Eberly (1994) theory of investment under uncertainty to include two capital goods that we interpret as physical and intangible capital. We present a stylized model, as our goal is to provide theoretical motivation for our empirical work, not to make a theoretical contribution. Wildasin (1984), Hayashi and Inoue (1991), and others already provide theories of investment in multiple capital goods. We first present the model's assumptions and predictions, and then we discuss them.

2.1. Model assumptions and empirical implications

The model features an infinitely lived, perfectly competitive firm i that holds K_{it}^{phy} units of physical capital and K_{it}^{int} units of intangible capital at time t . The firm's total capital is defined as $K^{tot} = K^{phy} + K^{int}$. At each instant t , the firm chooses the investment rates I^{phy} and I^{int} that maximize firm value V_{it} :

$$V_{it} = \max_{I_{i,t+s}^{phy}, I_{i,t+s}^{int}} \int_0^\infty E_t[\Pi(K_{i,t+s}^{tot}, \varepsilon_{i,t+s}) - c_i^{phy}(I_{i,t+s}^{phy}, K_{i,t+s}^{tot}, p_{i,t+s}^{phy}) - c_i^{int}(I_{i,t+s}^{int}, K_{i,t+s}^{tot}, p_{i,t+s}^{int})] e^{-rs} ds, \quad (1)$$

subject to

$$dK^m = (I^m - \delta K^m)dt, \quad m = phy, int. \quad (2)$$

Both types of capital depreciate at the same rate δ . The profit function Π depends on a shock ε and is assumed linearly homogenous in K^{tot} . The two investment-cost functions c equal

$$c_i^m(I^m, K^{tot}, p^m) = p^m I^m + K^{tot} \left[\zeta_i^m \frac{I^m}{K^{tot}} + \frac{\gamma_i^m}{2} \left(\frac{I^m}{K^{tot}} \right)^2 \right], \quad m = phy, int, \quad (3)$$

where $\gamma_i > 0$. The first term denotes the direct purchase or sale cost of investment, with each new unit of capital

costing p^m . The second term equals the cost of adjusting the stock of capital type m . Capital prices p_{it}^{phy} and p_{it}^{int} , along with profitability shock ε_{it} , fluctuate over time according to a general stochastic diffusion process:

$$dy_{it} = \mu(y_{it})dt + \Sigma(y_{it})dB_{it}, \quad (4)$$

where $y_{it} = [\varepsilon_{it} \ p_{it}^{phy} \ p_{it}^{int}]'$.

We have four main predictions. All proofs are in Appendix A.

Prediction 1. Physical and intangible capital share the same marginal q . Marginal q equals average q , the ratio of firm value to its total capital stock:

$$\frac{\partial V_{it}}{\partial K_{it}^{phy}} = \frac{\partial V_{it}}{\partial K_{it}^{int}} = \frac{\partial V_{it}}{\partial K_{it}^{tot}} = \frac{V_{it}}{K_{it}^{tot}} \equiv q^{tot}(\varepsilon_{it}, p_{it}^{phy}, p_{it}^{int}). \quad (5)$$

Marginal q equals $\partial V / \partial K$ and measures the benefit of adding an incremental unit of capital (either physical or intangible) to the firm. Marginal q equals average q , because we assume constant returns to scale, perfect competition, and perfect substitutes in production and depreciation. This prediction provides a rationale for measuring Tobin's q as q^{tot} , firm value divided by K^{tot} , the sum of physical and intangible capital. The value of q^{tot} depends endogenously on the shock ε and the two capital prices.

The firm chooses its optimal investment rates by equating their marginal q and their marginal cost of investment. Applying this condition to Eq. (3) yields Prediction 2.

Prediction 2. The firm's optimal physical and intangible investment rates follow

$$I_{it}^{phy} = \frac{I_{it}^{phy}}{K_{it}^{tot}} = \frac{1}{\gamma_i^{phy}} (q_{it}^{tot} - \zeta_i^{phy} - p_{it}^{phy}) \quad (6)$$

$$I_{it}^{int} = \frac{I_{it}^{int}}{K_{it}^{tot}} = \frac{1}{\gamma_i^{int}} (q_{it}^{tot} - \zeta_i^{int} - p_{it}^{int}). \quad (7)$$

Prediction 2 says that the physical and intangible investment rates, both scaled by total capital, vary with q^{tot} . One empirical implication is that physical and intangible investment rates should be correlated. The correlation might not be perfect, though, because adjustment-cost parameters might not be perfectly correlated across firms, and the prices p_{it}^{phy} and p_{it}^{int} might not be perfectly correlated either across firms or over time.

Predictions 3 and 4 follow immediately from Prediction 2 and form the basis of our empirical work. Consider a panel of firms indexed by i . We assume parameters γ^{phy} and γ^{int} are constant across firms, but other parameters and shocks can vary across firms. We assume the two capital prices p_{it}^m can be decomposed as $p_{it}^m = p_i^m + p_t^m$.

Prediction 3. In an OLS panel regression of I_{it}^{phy} on q_{it}^{tot} and firm and time fixed effects (FEs), the slope on q equals $1/\gamma^{phy}$. If the dependent variable is instead I_{it}^{int} , the q -slope equals $1/\gamma^{int}$. If the dependent variable is I_{it}^{tot} , the q -slope equals $1/\gamma^{phy} + 1/\gamma^{int}$. Any other regressors, such as free cash flow, should not enter significantly if added to any of these regressions.

Prediction 3 says that total q helps explain all three investment measures, and it shows that the OLS slopes identify the adjustment-cost parameters γ . The firm and

year FEs are needed to absorb the terms $-\zeta_i - p_{it}$ in Eqs. (6) and (7).

To our knowledge, Prediction 4 is new to the literature. It helps explain the investment literature's typical regression, which excludes intangible capital and instead scales investment and q by physical capital alone.

Prediction 4. Define $q_{it}^* = V_{it}/K_{it}^{phy}$ and $\iota_{it}^* = I_{it}^{phy}/K_{it}^{phy}$. In an OLS panel regression of ι_{it}^* on q_{it}^* and firm and time fixed effects, the slope coefficient is a downward-biased estimate of $1/\gamma^{phy}$, and the R^2 is lower than the R^2 from the regressions in Prediction 3.

According to our theory, this regression is misspecified, because the ratio $-K_{it}^{tot}/K_{it}^{phy}$ is part of the regression's disturbance and cannot be explained by the FEs. Its q -slope is downward-biased, meaning it produces upward-biased estimates of the adjustment-cost parameter γ^{phy} , because q_{it}^* depends on the ratio $K_{it}^{tot}/K_{it}^{phy}$, making the regressor negatively related to the disturbance.

At this point, we have imposed several restrictive assumptions. To help judge the model's empirical relevance, we establish one last prediction and use it as a consistency check in our empirical work. This last prediction links firms' use of intangible capital to their adjustment costs and q -slopes. If we impose the additional assumptions that physical and intangible capital have the same linear adjustment-cost parameters ($\zeta_i^{phy} = \zeta_i^{int}$) and purchase prices ($p_{it}^{phy} = p_{it}^{int}$), then

$$\lim_{t \rightarrow \infty} \frac{K_{it}^{int}}{K_{it}^{tot}} = \frac{\gamma^{phy}}{\gamma^{phy} + \gamma^{int}} = \frac{\beta^{int}}{\beta^{int} + \beta^{phy}}, \quad (8)$$

where β^{int} and β^{phy} are the Prediction 3 slopes of ι^{int} and ι^{phy} , respectively, on q^{tot} . Intuitively, if physical and intangible capital are identical except for their adjustment-cost parameters γ , then a firm holds relatively less intangible capital if intangible capital is costlier to adjust ($\gamma^{int} > \gamma^{phy}$). Section 5 performs a consistency check by comparing the Eq. (8) ratio of regression slopes across firms with different amounts of intangible capital.

2.2. Discussion

To summarize, our simple theory predicts that total q helps explain physical, intangible, and total investments when we scale them by the firm's total capital. It also illustrates how investment regressions can identify the convex part (γ) of capital adjustment costs. The theory also predicts that including intangible capital produces a better-specified investment regression and more accurate adjustment-cost estimates.

Next, we discuss the theory's assumptions and limitations. Overall, we argue that intangible capital fits well into the neoclassical framework.

Conceptually, spending on intangible assets qualifies as a capital investment, because it reduces current cash flow to increase future cash flow (Corrado, Hulten, and Sichel, 2005; 2009). Ample evidence exists that intangible investments increase firms' future profits, as our theory assumes. A large R&D literature (e.g., Lev and Sougiannis, 1996) shows that R&D investments increase firms' future profits. Recognizing this fact, the Bureau of Economic Analysis

(BEA) began capitalizing R&D in satellite accounts in 1994 and in core National Income and Product Accounts (NIPA) in 2013. A large marketing literature (e.g., Aaker, 1991; Srivastava, Shervani, and Fahey, 1997) shows that firms with stronger brands earn higher profits and are worth more. More generally, Eisfeldt and Papanikolaou (2013) show that firms using more organization capital are more productive after accounting for physical capital and labor. Even though a firm does not own its workers, employee training builds the firm's human capital, because training is costly and increases the firm's future profits.

While employee training and brand building can entail relatively low risk, investments such as R&D projects are highly risky and sometimes fail completely. The same is true for physical investments, though. Our theory is designed to handle investments with risky payoffs, so payoff risk is no reason to exclude intangible capital from the neoclassical theory. In addition to payoff risk, firms face depreciation risk. Our theory assumes a constant depreciation rate for intangible capital, whereas the true rate is likely random. For example, writing off a large portion of knowledge capital could be appropriate when a firm narrowly loses a patent race. Physical capital's true depreciation rate is also likely random, however. For example, an unexpected product-market change could make a machine obsolete. Again, no conceptual difference exists between physical and intangible capital here, although there could be a difference of degree.

When researchers test investment theories, they usually measure investment as CAPX and capital as PP&E. These two measures add together physical assets that are conceptually very different from each other, such as timberland, medical equipment, oil reserves, computers, buildings, and so on. By using such measures, researchers implicitly treat these physical assets as perfect substitutes. Similarly, our theory adds together many different types of intangible assets into K^{int} , and then it assumes the firm's profits depend on K^{tot} , the sum of physical and intangible capital. We therefore treat all assets as perfect substitutes in producing profits, although we do allow them to have potentially different adjustment costs. In our opinion, a natural first step is to treat intangible capital the same way researchers for decades have treated physical capital. In reality, physical and intangible capital could be complements, not substitutes. One might therefore expect our empirical measures, which simply add together all capital, to produce poor results. We find the opposite, which is somewhat surprising and suggests that our simple model provides a useful approximation of reality.

The theory highlights an important limitation of investment regressions. Whited (1994) and Erickson and Whited (2000) explain that investment regressions cannot identify the level of adjustment costs. For example, our theory predicts that the linear adjustment-cost parameters ζ are not separately identified from firm-specific capital prices p . The investment regression identifies only the quadratic adjustment-cost parameters γ , meaning the investment regression can identify only the convex component of adjustment costs. This convex component is interesting, however, as it determines how investment responds to investment opportunities.

3. Firm-level data

Our sample includes all Compustat firms except regulated utilities (Standard Industrial Classification codes 4900–4999), financial firms (6000–6999), and firms categorized as public service, international affairs, or non-operating establishments (9000+). We also exclude firms with missing or non-positive book value of assets or sales and firms with less than \$5 million in physical capital, as is standard in the literature. We use data from 1975 to 2011, although we use earlier data to estimate firms' intangible capital. Our sample starts in 1975, because this is the first year that the Federal Accounting Standards Board (FASB) requires firms to report R&D. We winsorize all regression variables at the 1% level to remove outliers.

3.1. Tobin's q

Guided by our theory, we measure total q by scaling firm value by the sum of physical and intangible capital:

$$q_{it}^{tot} = \frac{V_{it}}{K_{it}^{phy} + K_{it}^{int}}. \quad (9)$$

We measure the replacement cost of physical capital, K_{it}^{phy} , as the book value of property, plant, and equipment (Compustat item *ppegt*). Section 3.2 defines our measure of K_{it}^{int} , the replacement cost of intangible capital. We measure the firm's market value V as the market value of outstanding equity (Compustat items *prcc_f* times *csho*), plus the book value of debt (Compustat items *dltt* + *dlc*), minus the firm's current assets (Compustat item *act*), which include cash, inventory, and marketable securities.

For comparison, we examine the literature's standard Tobin's q measure used by Fazzari, Hubbard, and Petersen (1988), Erickson and Whited (2012), and many others:

$$q_{it}^* = \frac{V_{it}}{K_{it}^{phy}}. \quad (10)$$

Erickson and Whited (2006), Erickson and Whited (2012) compare several alternate Tobin's q measures, including the market-to-book-assets ratio, and they find that q^* best explains investment. The correlation between q^* and q^{tot} is 0.82.

3.2. Intangible capital

We briefly review the US accounting rules for intangible capital before defining our measure.³ The accounting rules depend on whether the firm creates the intangible asset internally or purchases it externally.

Intangible assets created within a firm are expensed on the income statement and almost never appear as assets on the balance sheet. For example, a firm's spending to develop knowledge, patents, or software is expensed as R&D. Advertising to build brand capital is a selling expense within SG&A. Employee training to build human capital is

a general or administrative expense within SG&A. There are a few exceptions, in which internally created intangibles are capitalized on the balance sheet, but these are small in magnitude.⁴

When a firm purchases an intangible asset externally, for example, by acquiring another firm, the firm typically capitalizes the asset on the balance sheet as part of Intangible Assets, which equals the sum of Goodwill and Other Intangible Assets. The asset is booked in Other Intangible Assets if the acquired asset is separately identifiable, such as a patent, software, or client list. Acquired assets that are not separately identifiable, such as human capital, are in Goodwill. When an intangible asset becomes impaired, firms are required to write down its book value.

We define the replacement cost of intangible capital, denoted K_{it}^{int} , to be the sum of the firm's externally purchased and internally created intangible capital. We measure externally purchased intangible capital as Intangible Assets from the balance sheet (Compustat item *intan*). We set this value to zero if missing. We keep Goodwill in Intangible Assets in our main analysis, because Goodwill does include the fair cost of acquiring intangible assets that are not separately identifiable. Because Goodwill can be contaminated by non-intangibles, such as a market premium for physical assets, we also try excluding Goodwill from external intangibles and show that our results are almost unchanged (Section 7). Our mean (median) firm purchases only 19% (3%) of its intangible capital externally, meaning the vast majority of firms' intangible assets are missing from their balance sheets. There are important outliers, however. For example, 41% of Google's intangible capital in 2013 had been purchased externally. Including these externally purchased intangibles is an innovation in our measure relative to those in the literature.

Measuring the replacement cost of internally created intangible assets is difficult, as they appear nowhere on the balance sheet. Fortunately, we can construct a proxy by accumulating past intangible investments, as reported on firms' income statements. We define the stock of internal intangible capital as the sum of knowledge capital and organization capital.

A firm develops knowledge capital by spending on R&D. We estimate a firm's knowledge capital by accumulating past R&D spending using the perpetual inventory method:

$$G_{it} = (1 - \delta_{R\&D})G_{i,t-1} + R\&D_{it}, \quad (11)$$

where G_{it} is the end-of-period stock of knowledge capital, $\delta_{R\&D}$ is its depreciation rate, and $R\&D_{it}$ is real expenditures on R&D during the year. The BEA uses a similar method to capitalize R&D, as do practitioners when valuing companies (Damodaran, 1999, 2001). For $\delta_{R\&D}$, we use the

³ Chapter 12 in Kieso, Weygandt, and Warfield (2010) provides a useful summary of the accounting rules for intangible assets. The authors also provide references to relevant FASB codifications.

⁴ Our measure captures these exceptions via balance sheet Intangible Assets. Firms capitalize the legal costs, consulting fees, and registration fees incurred when developing a patent or trademark. A firm can start capitalizing software spending only after the product reaches technological feasibility (for externally sold software) or reaches the coding phase (for internally used software). The resulting software asset is part of Other Intangible Assets (*intano*) in Compustat.

BEA's industry-specific R&D depreciation rates.⁵ We measure annual R&D using the Compustat variable *xrd*. We use Compustat data back to 1950 to compute Eq. (11), but our regressions include only observations starting in 1975. Starting in 1977, we set R&D to zero when missing, following Lev and Radhakrishnan (2005) and others.⁶

One challenge in applying the perpetual inventory method in Eq. (11) is choosing a value for G_{i0} , the capital stock in the firm's first non-missing Compustat record, which usually coincides with the initial public offering (IPO). We estimate G_{i0} using data on the firm's founding year, R&D spending in its first Compustat record, and average pre-IPO R&D growth rates. With these data, we estimate the firm's R&D spending in each year between its founding and appearance in Compustat. We apply a similar approach to SG&A. Appendix B provides additional details. Section 7 shows that a simpler measure assuming $G_{i0} = 0$ produces an even stronger investment- q relation than our main measure. We consider that simpler measure a reasonable alternate proxy for investment opportunities.

Next, we measure the stock of organization capital by accumulating a fraction of past SG&A spending using the perpetual inventory method, as in Eq. (11). The logic is that at least part of SG&A represents an investment in organization capital through advertising, spending on distribution systems, employee training, and payments to strategy consultants. We follow Hulten and Hao (2008), Eisfeldt and Papanikolaou (2014), and Zhang (2014) in counting only 30% of SG&A spending as an investment in intangible capital. We interpret the remaining 70% as operating costs that support the current period's profits. Section 7 shows that our conclusions still go through, albeit with smaller magnitudes, if we use values other than 30%. We follow Falato, Kadyrzhanova, and Sim (2013) in using a depreciation rate of $\delta_{SG\&A} = 20\%$, and in Section 7 we show that our conclusions are robust to alternate depreciation rates.

Measuring SG&A from Compustat data is not trivial. Companies typically report SG&A and R&D separately. Compustat, however, almost always adds them together in a variable misleadingly labeled "Selling, General and Administrative Expense" (item *xsga*). We must therefore subtract *xrd* from *xsga* to isolate the SG&A that companies report. Appendix B provides additional details.

Our measure of internally created organization capital is almost identical to that of Eisfeldt and Papanikolaou (2012), Eisfeldt and Papanikolaou (2013), and

Eisfeldt and Papanikolaou (2014). They validate the measure in several ways. They show a positive correlation between firms' use of organization capital and the Bloom and Van Reenen (2007) managerial quality score. This score is associated with higher firm profitability, production efficiency, and productivity of information technology (IT) (Bloom, Sadun, and Van Reenen, 2010). Eisfeldt and Papanikolaou (2013) show that firms using more organization capital are more productive after accounting for physical capital and labor, spend more on IT, and employ higher-skilled workers. They show that firms with more organization capital list the loss of key personnel as a risk factor more often in their 10-K filings. Practitioners also use our approach. A popular textbook on value investing recommends capitalizing SG&A to measure assets missing from the balance sheet (Greenwald, Kahn, Sonkin, and Van Biema, 2004).

Our measure of intangible capital has the benefit of being easily computed for the full Compustat sample. The measure has limitations, however, as discussed in Section 2.2. Section 4.2 addresses concerns about measurement error bias, and Section 7 shows that our conclusions are robust to several alternate ways of measuring intangible capital. Overall, we believe, and the data confirm, that an imperfect proxy for intangible capital is better than setting it to zero.

3.3. Investment

Guided by our theory, we measure the firm's physical, intangible, and total investment rates as

$$i_{it}^{phy} = \frac{I_{it}^{phy}}{K_{i,t-1}^{tot}}, \quad i_{it}^{int} = \frac{I_{it}^{int}}{K_{i,t-1}^{tot}}, \quad i_{it}^{tot} = i_{it}^{phy} + i_{it}^{int}. \quad (12)$$

We measure physical investment I^{phy} as capital expenditures (Compustat item *capx*), and we measure intangible investment, I^{int} , as R&D + $(0.3 \times SG\&A)$. This definition assumes 30% of SG&A represents an investment, as we assume when estimating capital stocks. For comparison, we examine the literature's standard physical investment measure, denoted i^* in our theory:

$$i_{it}^* = \frac{I_{it}^{phy}}{K_{i,t-1}^{phy}}. \quad (13)$$

The correlation between i_{it}^{phy} and i_{it}^* is 0.83.

3.4. Cash flow

Erickson and Whited (2012), Almeida and Campello (2007), and others measure free cash flow as

$$c_{it}^* = \frac{IB_{it} + DP_{it}}{K_{i,t-1}^{phy}}, \quad (14)$$

where *IB* is income before extraordinary items and *DP* is depreciation expense. The measure c^* is the pre-depreciation free cash flow available for physical investment or distribution to shareholders. One shortcoming of c^* is that it treats R&D and SG&A as operating expenses, not investments. In addition to the standard measure c^* ,

⁵ The BEA's R&D depreciation rates are from the analysis of Li (2012). The depreciation rates range from 10% in the pharmaceutical industry to 40% for computers and peripheral equipment. Following the BEA's guidance, we use a depreciation rate of 15% for industries not in Li's Table 4. Our results are virtually unchanged if we set $\delta_{R\&D}$ equal to 10%, 15%, or 20% for all industries (Table 9).

⁶ We start in 1977 to give firms two years to comply with FASB's 1975 R&D reporting requirement. If we see a firm with R&D equal to zero or missing in 1977, we assume the firm was typically not an R&D spender before 1977, so we set any missing R&D values before 1977 to zero. Otherwise, before 1977, we either interpolate between the most recent non-missing R&D values (if such observations exist) or we use the method in Appendix A (if those observations do not exist). Starting in 1977, we make exceptions in cases in which the firm's assets are also missing. These are likely years when the firm was privately owned. In such cases, we interpolate R&D values using the nearest non-missing values.

Table 1

Summary statistics.

Statistics are based on the sample of Compustat firms from 1975 to 2011. The physical capital stock, K^{phy} , is measured as property, plant, and equipment (PP&E). We estimate the intangible capital stock, K^{int} , by applying the perpetual inventory method to firms' intangible investments, defined as research and development (R&D) and $0.3 \times$ selling, general, and administrative (SG&A) spending. We then add in firms' balance sheet intangibles. Intangible intensity equals $K^{int} / (K^{int} + K^{phy})$. Knowledge capital is the part of intangible capital that comes from R&D. The denominator for all new measures is $K^{int} + K^{phy}$. The denominator for all standard measures is K^{phy} . The numerator for both q variables is the market value of equity plus the book value of debt minus current assets. The numerator for q^{phy} is capital expenditure (CAPX), and the numerator for q^{int} is R&D + $(0.3 \times \text{SG\&A})$. Total investment $i^{tot} = i^{phy} + i^{int}$. The numerator for standard cash flow is income before extraordinary items plus depreciation expenses. The numerator for total cash flow is the same but adds back intangible investment net a tax adjustment.

Variable	Mean	Median	Standard deviation	Skewness
Intangible capital stock (millions of dollars)	427	41.7	1990	11.6
Physical capital stock (millions of dollars)	1237	77.9	6691	16.5
Intangible intensity	0.43	0.45	0.27	−0.01
Knowledge capital/intangible capital	0.24	0.01	0.37	1.65
New measures				
Total q (q^{tot})	1.11	0.57	1.91	3.76
Physical investment (i^{phy})	0.10	0.06	0.14	3.47
Intangible investment (i^{int})	0.11	0.09	0.11	1.92
Total investment (i^{tot})	0.21	0.16	0.18	2.61
Total cash flow (c^{tot})	0.16	0.15	0.19	0.52
Standard measures				
Standard q (q^*)	3.14	0.93	7.22	4.41
CAPX/PPE (i^*)	0.19	0.11	0.24	3.52
Standard cash flow (c^*)	0.15	0.16	0.62	−1.63

we use an alternate cash flow measure that recognizes R&D and part of SG&A as investments. We add intangible investments back into the free cash flow so that we measure the profits available for total, not just physical, investment:

$$c_{it}^{tot} = \frac{IB_{it} + DP_{it} + i_{it}^{int} (1 - \kappa)}{K_{i,t-1}^{phy} + K_{i,t-1}^{int}}. \quad (15)$$

Lev and Sougiannis (1996) similarly adjust earnings for intangible investments, as do practitioners (Damodaran, 1999, 2001). Because accounting rules allow firms to expense intangible investments, the effective cost of a dollar of intangible capital is only $(1 - \kappa)$, where κ is the marginal tax rate. When available, we use simulated marginal tax rates from Graham (1996). Otherwise, we assume a marginal tax rate of 30%, which is close to the mean tax rate in the sample. The correlation between c^{tot} and c^* is 0.77.

3.5. Summary statistics

Table 1 contains summary statistics. We define intangible intensity as a firm's ratio of intangible to total capital, at replacement cost. The mean (median) intangible intensity is 43% (45%), so almost half of capital is intangible in our typical firm-year. Knowledge capital makes up only 24% of intangible capital on average, so organization capital makes up 76%. The median firm has almost no knowledge capital, as almost half of firms report no R&D. The average q^{tot} is mechanically smaller than q^* , because its denominator is larger. The gap is dramatic in some cases. For example, Google's q^* is 10.1 in 2013, but its q^{tot} is only 3.2. Researchers sometimes discard q observations exceeding 10, arguing they are unrealistically large. Total q exceeds 10 in only 1% of observations, compared with 7% for standard q , suggesting total q is a more reliable measure. The standard

deviation of q^{tot} is 74% lower than for q^* . The standard deviation scaled by its mean is also lower. The average physical and intangible investment rates are roughly equal, but physical investment is more volatile and right-skewed.

Fig. 1 shows that the average intangible intensity has increased over time, especially in the 1990s. The figure also shows that high-tech and health firms are heavy users of intangible capital and that manufacturing firms use less. Somewhat surprisingly, even manufacturing firms' capital is 30–34% intangible on average.

4. Full-sample results

In this section, we test the theory's predictions in our full sample. Section 5 compares results across subsamples. We begin with the classic OLS panel regressions of Fazzari, Hubbard, and Petersen (1988). We then correct for measurement error bias in Section 4.2.

4.1. OLS results and co-movement in investment

Table 2 contains results from OLS regressions of investment on lagged q and firm and year fixed effects. The columns compare different investment measures. For now, we focus on R^2 values, because the regression coefficients suffer from measurement error bias. This bias is especially severe for cash flow coefficients (Abel, 2016; Erickson and Whited, 2000), so we exclude cash flow until Section 4.2.

Taken literally, the theory predicts an R^2 of 100% in Panel A when we measure investment as i^{phy} , i^{int} , or i^{tot} . We find R^2 values that are well below 100%. One potential explanation is that we measure q with error, an issue we address in Section 4.2. Another is that slopes vary across firms or that shocks hit firms' marginal adjustment cost functions. Our theory's prediction holds better for intangible investment ($R^2 = 27.9\%$) than for physical

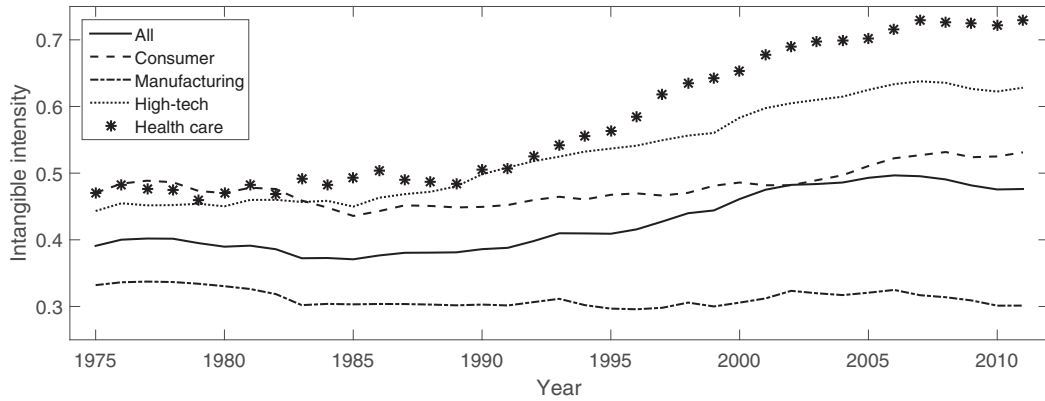


Fig. 1. Capital intangibility over time. This figure plots the mean intangible capital intensity over time, both for our full sample and within industries. Intangible intensity equals $K^{int}/(K^{int} + K^{phy})$, the firm's stock of intangible capital divided by its total stock of capital. We use the Fama and French five-industry definition and exclude industry "Other".

Table 2

Ordinary least squares results.

Results are from OLS panel regressions of investment on lagged Tobin's q and firm and year fixed effects. Each column uses a different investment measure. Physical investment (ι^{phy}) equals capital expenditure (CAPX) scaled by total capital ($K^{tot} = K^{phy} + K^{int}$). Intangible investment (ι^{int}) equals research and development (R&D) + $0.3 \times$ selling, general and administrative (SG&A) expense, scaled by K^{tot} . Total investment equals $\iota^{phy} + \iota^{int}$. R&D investment equals R&D scaled by total capital. The R&D column excludes observations with missing R&D. The investment measure in the final column is CAPX divided by property, plant and equipment (PP&E). Panel A shows regressions on total q , denoted q^{tot} . Panel B shows regressions on standard q , denoted q^* . The numerator for both q variables is the market value of equity plus the book value of debt minus current assets. The denominator for q^{tot} is K^{tot} . The denominator for q^* is K^{phy} . Standard errors clustered by firm are in parentheses. We report the within-firm R^2 . Panel C tests whether the R^2 values in Panels A and B are different, taking into account the correlation across regressions and again clustering by firm. Data are from Compustat from 1975 to 2011.

	Investment scaled by total capital (K^{tot})				
	Physical (ι^{phy})	Intangible (ι^{int})	Total (ι^{tot})	R&D	CAPX/PPE (ι^*)
Panel A: Regressions with total q					
Total q	0.029	0.020	0.049	0.013	0.062
	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)
R^2	0.209	0.279	0.327	0.270	0.244
	(0.008)	(0.007)	(0.006)	(0.009)	(0.008)
Panel B: Regressions with standard q					
Standard q	0.006	0.005	0.011	0.003	0.017
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
R^2	0.139	0.266	0.250	0.250	0.233
	(0.009)	(0.008)	(0.007)	(0.010)	(0.008)
Panel C: Difference in R^2 (Panel A–Panel B)					
ΔR^2	0.070	0.013	0.077	0.020	0.011
	(0.003)	(0.004)	(0.003)	(0.005)	(0.003)
Number of observations	141,800	141,800	141,800	75,426	141,800

investment ($R^2 = 20.9\%$), and it holds better still for total investment ($R^2 = 32.7\%$). We also check that this result holds for the portion of intangible investment coming from R&D, because the portion from SG&A is measured with more error. When we measure investment as R&D scaled by total capital, we find an R^2 of 27.0%, which is similar to the 27.9% R^2 from our main intangible investment measure, ι^{int} .

Our theory predicts a lower R^2 for the literature's usual regression of CAPX/PPE on standard q , shown in the last column of Panel B. The R^2 here is low (23.3%) relative to all the R^2 values in Panel A, with one exception: Standard q explains standard investment slightly better than total q explains our new physical investment measure, ι^{phy} . For ι^{phy} , measurement error in intangible capital could be offsetting any improvements from including intangible capital in the denominator of q .

One interesting implication of our theory is that physical and intangible investment should co-move strongly within firms, because the two capital types have the same marginal productivity and, hence, the same marginal q . We find strong co-movement in the data: ι^{phy} and ι^{int} have a 31% correlation after we remove firm and time fixed effects from both. According to the theory, this co-movement should decrease if we remove the effects of total q , for example, by isolating the residuals for ι^{phy} and ι^{int} from Panel A. The correlation between these two regressions' residuals is lower (17%), consistent with the theory. This remaining correlation could just be an artifact of measurement error in total q .

Throughout the corporate finance literature, researchers use Tobin's q to proxy for firms' investment opportunities. The R^2 values in Table 2 help us judge how well these proxies work and, in particular, whether total q or the

literature's standard q measure is the better proxy for investment opportunities. Panel B shows how well standard q explains the five investment measures, and Panel C tests whether total q or standard q delivers a higher R^2 .⁷ For all five investment measures, total q delivers a larger R^2 value than standard q . The improvement in R^2 ranges from 1 to 8 percentage points, or from 5% to 50%. Some of the improvements are modest in magnitude, but statistical significance in Panel C is high, with t -statistics ranging from 3.4 to 25.

It is tempting to run a horse race by including total and standard q in the same regression. Because both variables proxy for q with error, their resulting slopes would be biased in an unknown direction, making the results difficult to interpret (Klepper and Leamer, 1984). For this reason, we do not tabulate results from such a horse race. We simply note that regressing either ι^{phy} or ι^{tot} on both q proxies produces a positive and highly significant slope on q^{tot} but a negative and less significant slope on q^* . For ι^{int} and ι^* , both q variables have a significantly positive slope, but the slope on q^{tot} is much larger in magnitude.

Outside the investment literature, it is popular to measure Tobin's q as the firm's market value scaled by its book value of assets. Like Erickson and Whited (2006), Erickson and Whited (2012), we find that these market-to-book-assets ratios are especially poor proxies for investment opportunities. They produce lower R^2 values than both standard and total q no matter how we measure investment (Online Appendix, Table A1).

To summarize, total q explains intangible investment slightly better than physical investment in our full sample, and it explains total investment even better. As our theory predicts, physical and intangible investment co-move strongly within firms, because they share the same q . This result suggests strong co-movement between physical and intangible capital's marginal productivities. Judging by these results, the neoclassical theory of investment is just as relevant for intangible capital as it is for physical capital. We also show that total q is a superior proxy for investment opportunities no matter how investment is measured.

4.2. Bias-corrected results

According to our theory, total q is better than standard q at approximating the true, unobservable q . We recognize, however, that total q is still a noisy proxy. For one, we measure intangible capital with error. Also, Tobin's q measures average q , but investment depends on marginal q in theory. Average q equals marginal q in our simple theory, but, to the extent that reality departs from this theory, average q measures marginal q with error.⁸

⁷ Throughout, we conduct inference on R^2 values using influence functions (Newey and McFadden, 1994). In a regression $y = \beta x + \epsilon$, this approach takes into account the estimation error in β , $\text{var}(y)$, and $\text{var}(x)$. We cluster by firm, which accounts for autocorrelation both within and across regressions.

⁸ Gala (2014) measures the differences between marginal and average q .

Because we have only a proxy for q , all the OLS slopes from Section 4.1 suffer from measurement error bias. We now estimate the previous models while correcting this bias using the Erickson, Jiang, and Whited (2014) higher-order cumulant estimator.⁹ The cumulant estimator provides unbiased estimates of β in the following classical errors-in-variables model:

$$\iota_{it} = a_i + q_{it}\beta + z_{it}\alpha + u_{it} \quad (16)$$

$$p_{it} = \gamma + q_{it} + \varepsilon_{it}, \quad (17)$$

where p is a noisy proxy for the true, unobservable q and z is a vector of perfectly measured control variables. The cumulant estimator's main identifying assumptions are that p has nonzero skewness, $\beta \neq 0$, and u and ε are independent of q , z , and each other.

Because the cumulant estimator corrects for measurement error, why is a new q proxy with less measurement error needed? The reason is that, by ignoring intangibles, the literature's standard physical investment and q proxies, ι^* and q^* , are both mismeasured, and the measurement error is multiplicative, not additive. The measurement error in ι^* and q^* comes from the omission of intangibles. Because that same error is in both variables, their measurement errors are correlated with each other, violating the cumulant estimator's assumption that u and ε are independent.¹⁰ The measurement error is multiplicative, because changing the variables' denominators from physical to total capital requires multiplying them both by K^{phy}/K^{tot} . The multiplicative error makes both variables' errors depend on the true q , violating the cumulant estimator's assumption that u and ε are independent of q . We cannot solve the problem by regressing total investment on the standard q measure (ι^{tot} on q^*), because measurement error in q^* is still multiplicative and, hence, a function of q , again violating the cumulant estimator's assumption that ε is independent of q .¹¹

We perform a simple horse race to illustrate that the cumulant estimator on its own cannot correct for the measurement error in the standard q measure. Using the cumulant estimator, we regress ι^{tot} on q^* , and then regress

⁹ The cumulant estimator supersedes the Erickson and Whited (2002) higher-order moment estimator. Cumulants are polynomials of moments. The estimator is a generalized method of moments (GMM) estimator with moments equal to higher-order cumulants of investment and q . Compared with the Erickson and Whited (2002) estimator, the cumulant estimator has better finite-sample properties and a closed-form solution, which makes numerical implementation easier and more reliable. We use the third-order cumulant estimator, which dominates the fourth-order estimator in the estimation of τ^2 (Erickson, Jiang, and Whited, 2014; Erickson and Whited, 2012). Results are similar using the fourth-order cumulant estimator (Online Appendix).

¹⁰ To see this, assume that the world behaves according to $\iota_{it}^{\iota\alpha} = q_{it}\beta$, where q_{it} is the unobservable, true q ; that our empirical proxy $q_{it}^{\iota\alpha} = q_{it} + \varepsilon_{it}$, where ε_{it} is independently distributed; and that we mistakenly estimate the errors-in-variables model using the standard measures: $\iota_{it}^{\iota\alpha} = q_{it}\beta + u_{it}$ and $q_{it}^{\iota\alpha} = q_{it} + \varepsilon_{it}$. One can prove that $u_{it} = q_{it}(A_{it}B_{it}\beta - \beta)$ and $\varepsilon_{it} = q_{it}(B_{it} - 1) + B_{it}\varepsilon_{it}$, where $B_{it} = K_{it}^{\iota\alpha}/K_{it}^{phy}$ and $A_{it} = I_{it}^{phy}/I_{it}^{\iota\alpha}$. Because u_{it} and ε_{it} both depend on $q_{it}B_{it}$, they are not independent of q or each other.

¹¹ To see this, suppose that the assumptions in footnote 10 hold, except we instead estimate the errors-in-variables model $\iota_{it}^{\iota\alpha} = q_{it}\beta + u_{it}$, and $q_{it}^{\iota\alpha} = q_{it} + \varepsilon_{it}$. One can prove that $\varepsilon_{it} = q_{it}(B_{it} - 1) + \varepsilon_{it}B_{it}$, so ε_{it} and q_{it} are not independent of each other.

Table 3

Bias-corrected results.

Results are from regressions of investment on lagged Tobin's q , firm fixed effects, and (in Panel B) contemporaneous cash flow, all estimated using the cumulant estimator. Each column uses a different investment measure as defined in Table 2. Total (standard) q equals the firm's market value scaled by K^{tot} (K^{phy}). The numerator for standard cash flow is income before extraordinary items plus depreciation expenses. The numerator for total cash flow is the same but adds back intangible investment net a tax adjustment. Total cash flow is scaled by K^{tot} ; standard cash flow, by K^{phy} . ρ^2 is the within-firm R^2 from a hypothetical regression of investment on true q , and τ^2 is the within-firm R^2 from a hypothetical regression of our q proxy on true q . For comparison, the table also shows the ordinary least squares (OLS) R^2 values from Table 2. Standard errors clustered by firm are in parentheses. Data are from Compustat from 1975 to 2011.

	Investment scaled by total capital (K^{tot})				CAPX/PPE (ι^*)
	Physical (ι^{phy})	Intangible (ι^{int})	Total (ι^{tot})	R&D	
Panel A: Regressions without cash flow					
Total q (q^{tot})	0.070 (0.001)	0.037 (0.001)	0.086 (0.001)	0.023 (0.000)	
Standard q (q^*)					0.036 (0.001)
OLS R^2	0.209 (0.008)	0.279 (0.007)	0.327 (0.006)	0.270 (0.009)	0.233 (0.008)
ρ^2	0.358 (0.008)	0.392 (0.008)	0.423 (0.008)	0.376 (0.011)	0.372 (0.008)
τ^2	0.437 (0.009)	0.559 (0.012)	0.597 (0.010)	0.593 (0.016)	0.492 (0.010)
Panel B: Regressions with cash flow					
Total q (q^{tot})	0.069 (0.001)	0.038 (0.001)	0.086 (0.002)	0.024 (0.001)	
Total cash flow (c^{tot})	0.024 (0.008)	0.050 (0.004)	0.140 (0.009)	0.000 (0.004)	
Standard q (q^*)					0.035 (0.001)
Standard cash flow (c^*)					0.015 (0.004)
OLS R^2	0.235 (0.008)	0.326 (0.007)	0.374 (0.006)	0.281 (0.009)	0.233 (0.008)
ρ^2	0.361 (0.008)	0.447 (0.009)	0.481 (0.008)	0.405 (0.011)	0.371 (0.008)
τ^2	0.435 (0.010)	0.502 (0.014)	0.544 (0.011)	0.568 (0.017)	0.494 (0.011)
Number of observations	141,800	141,800	141,800	75,426	141,800

ι^{tot} on q^{tot} . If the cumulant estimator could correct for the measurement error in q^* , then the two q proxies should produce similar q -slope estimates and ρ^2 values (defined below). Instead, we find that using total q produces a significantly higher q -slope (0.086 versus 0.023) and higher ρ^2 (0.423 versus 0.314). Results are in the Online Appendix.

Estimation results are in Table 3. Regarding the slopes on q , our estimates imply that intangible capital's convex adjustment costs are roughly twice as large as those for physical capital. According to our theory, the q -slopes measure the inverse capital adjustment-cost parameters γ^{phy} and γ^{int} . In Panel A, the 0.070 slope for ι^{phy} is roughly double the 0.037 slope for ι^{int} . We obtain a similar result after controlling for cash flow (Panel B) and also if we isolate the R&D component of intangible investment (Column 4). As we explain in Section 2.2, an important caveat is that our regressions can identify the convex component but not the overall level of adjustment costs.

This result helps support the Brown, Fazzari, and Petersen (2009, p. 160) conjecture that "R&D likely has high adjustment costs ..., possibly substantially higher than the adjustment costs for physical investment." Their argument is that R&D involves spending on highly skilled technology workers who are costly to hire, train, and replace. Griliches and Hausman (1986), Hall (2002), Himmelberg

and Petersen (1994), and Grabowski (1968) make similar arguments about R&D, and one could make similar arguments about human capital investments that are part of SG&A. Empirical evidence supporting these arguments is currently very limited.¹² An important implication of our result is that firms adjust more slowly to shocks to their investment opportunities as the economy shifts toward intangible capital.

Table 3 also changes how we view physical capital's adjustment costs. The last column in Panel A shows the literature's standard regression, which omits intangible capital. Prediction 4 in our theory states that this regression delivers a downward-biased estimate of $1/\gamma^{phy}$, i.e., a downward-biased q -slope. Consistent with this prediction, this standard regression delivers a q -slope of

¹² Bernstein and Nadiri (1989a,b) and Mohnen, Nadiri, and Prucha (1986) report slower adjustment speeds for R&D capital than physical capital in most but not all industries and for most but not all of their adjustment-cost measures. Bernstein and Nadiri (1989a,b) use data on 48 firms from 1965 to 1978 and 35 firms from 1959 to 1966, respectively. Mohnen, Nadiri, and Prucha (1986) use three countries' aggregate data from 1965 to 1977. Intangible capital was less prevalent during these years, and US firms were not required to report R&D until 1975. Li, Liu, and Xue (2014) estimate a structural model off the cross section of stock returns, and they find larger adjustment costs for R&D capital. By focusing on R&D capital, all these papers exclude organization capital.

0.036, roughly half as large as the 0.070 slope from the regression using ι^{phy} and scaling q by total capital. As we explain in Section 2.1, the typical regression delivers downward-biased slopes because the ratio of physical to total capital is an omitted variable that it positively related to the regressor and negatively related to the residual. This result helps resolve a puzzle in the investment literature. Researchers since Summers (1981) have argued that investment- q regressions produce implausibly small q -slopes, i.e., large adjustment costs. We find that physical capital's q -slopes are twice as large as previously believed, once one accounts for intangible capital.

In addition to delivering unbiased q -slopes, the cumulant estimator produces two useful test statistics. The first, ρ^2 , is the hypothetical R^2 from Eq. (16). Loosely speaking, ρ^2 indicates how well the true, unobservable q explains investment, with $\rho^2 = 1$ implying a perfect relation. Taken literally, our theory predicts $\rho^2 = 1$ even if we measure q with error. The second statistic, τ^2 , is the hypothetical R^2 from Eq. (17). It indicates how well our q proxy explains true q , with $\tau^2 = 1$ implying a perfect proxy.

Comparing the total-investment regression with the literature's typical regression, we find that including intangible capital produces a stronger investment- q relation (ρ^2 of 0.423 versus 0.372, a 14% increase) and a better proxy for Tobin's q (τ^2 of 0.597 versus 0.492, a 21% increase). On these dimensions, the classic q -theory fits the data better when we account for intangible capital. Model fit is still far from perfect, though: q explains less than half the variation in investment, and our total- q proxy explains less than 60% of the variation in true q .

Finally, we discuss the cash flow slopes shown in Panel B. Our simple theory predicts a zero cash flow slope for regressions using ι^{phy} , ι^{int} , and ι^{tot} . We find that physical investment has a significantly positive cash flow slope, contrary to the theory's prediction. We also find that including intangible capital affects the sensitivity of physical investment to cash flow. The physical investment-cash flow sensitivity is 60% higher (0.024 versus 0.015) when we compare the specification with ι^{phy} with the standard regression using CAPX/PPE.

Compared with physical investment, intangible investment appears roughly twice as sensitive to cash flow (slope of 0.050 versus 0.024). Intangible investment is the sum of its R&D and SG&A components. Which component is most important for producing the high investment-cash flow sensitivity? Column 4 of Panel B shows that R&D investment has a slope of zero on cash flow, consistent with the theory's prediction. SG&A investment must be highly sensitive to cash flow. Indeed, we find that SG&A investment has a cash flow slope of 0.115, which is more than double intangible investment's slope of 0.050 (Online Appendix).

One concern here is that measurement error in SG&A investment is biasing its cash flow slope upward. Compared with R&D, we measure SG&A investment with considerable error. Our c^{tot} measure is gross of SG&A investment, meaning we add back SG&A investment when computing it (Eq. (15)). Any measurement error in SG&A investment, therefore, appears mechanically in both c^{tot} and SG&A investment itself, biasing its cash flow slope up-

ward. For this reason, we view 0.115 as an upper bound for SG&A investment's cash flow sensitivity. We provide a lower bound in the Online Appendix by creating an alternate cash flow measure that is net of SG&A and, therefore, immune from this concern. SG&A investment has a statistically insignificant slope of 0.008 on this alternate cash flow measure. This 0.008 slope provides a lower bound for the true slope, because netting SG&A from cash flow pushes down the cash flow slope, and an economically meaningful cash flow measure should be gross of all investment, including SG&A investment. In sum, we can provide only a wide range for SG&A investment's cash flow slope (0.008–0.115), which implies a wide range in intangible investment's cash flow slope (0.012–0.050). Whether physical or intangible investment is more sensitive to cash flow is unclear.

Even absent these measurement challenges, interpreting the investment-cash flow sensitivity is notoriously difficult. Fazzari, Hubbard, and Petersen (1988) interpret it as evidence of financing constraints. In contrast, theories by Gomes (2001), Alt (2003), Cooper and Ejarque (2003), Hennessy and Whited (2007), Abel and Eberly (2011), and Gaurio and Rudanko (2014) predict an investment-cash flow sensitivity even in the absence of financing constraints. For example, decreasing returns to scale can make cash flow informative about marginal q , even after controlling for Tobin's (average) q . We simply conclude that physical investment is even more sensitive to cash flow than previously believed, R&D investment is insensitive to cash flow, and SG&A investment's cash flow sensitivity remains unclear. Without performing a full structural estimation, it is difficult to tell whether these cash flow results are driven by financing constraints, diseconomies of scale, or some other source.

5. Comparing subsamples

Next, we compare results across firms, industries, and years. Doing so allows us to test our theory and compare adjustment costs across subsamples. It also lets us check our main results' robustness across subsamples, which we discuss in Section 7.

We reestimate the previous models in subsamples formed using three variables. First, we sort firms each year into quartiles based on their lagged intangible intensity (Table 4). Second, we use the Fama and French five-industry definition to compare the manufacturing, consumer, high-tech, and health industries (Table 5). Third, we compare the early (1972–1995) and late (1996–2011) parts of our sample (Table 6). For each subsample, we estimate regressions using ι^{phy} , ι^{int} , and ι^{tot} , as well as the standard regression with CAPX/PPE.

5.1. Testing the theory in subsamples

The classic q theory, including the theory in this paper, fits the data better in settings with more intangible capital. We find this improved fit on three dimensions.

First, R^2 values increase dramatically when moving from the lowest to highest intangible quartile (Table 4,

Table 4

Comparing firms with different amounts of intangible capital.

This table shows results from subsamples formed based on yearly quartiles of intangible intensity, which equals the ratio of a firm's intangible to total capital. The first row show each quartile's mean intangible intensity. Results are from regressions of investment on lagged q , firm fixed effects, and (in Panel E) contemporaneous cash flow. Slopes on q and cash flow, as well as ρ^2 and τ^2 values, are from the cumulant estimator. R^2 is from the ordinary least squares (OLS) estimator that includes year fixed effects. Specifications 1–3 use physical (i^{phy}), intangible (i^{int}), and total investment (i^{tot}), respectively, along with total q , all of which are scaled by total capital. Specification 4 uses standard investment ($i^* = \text{capital expenditure (CAPX)} / \text{property plant and equipment (PPE)}$) and standard q (q^*), which is scaled by physical capital. The last row in Panel A shows the ratio of the Specification 2 q -slope to the sum of slopes from Specifications 1 and 2. We conduct inference using the delta method. Specifications 5–8 in Panel E add standard cash flow (c^*) and total cash flow (c^{tot}), defined in Table 3. Standard errors clustered by firm are in parentheses. We use influence functions to conduct inference for ρ^2 and τ^2 .

Specification	Quartile 1	Quartile 2	Quartile 3	Quartile 4	Quartile 4 – 1	
					Difference	Standard error
Intangible intensity	8%	33%	56%	76%		
Panel A: Slopes on q						
(1) i^{phy} on q^{tot}	0.095	0.081	0.063	0.050	–0.045	(0.004)
(2) i^{int} on q^{tot}	0.027	0.032	0.035	0.038	0.011	(0.004)
(3) i^{tot} on q^{tot}	0.101	0.097	0.086	0.074	–0.027	(0.004)
(4) CAPX/PPE on q^*	0.065	0.052	0.035	0.033	–0.032	(0.007)
$\beta^{int} / (\beta^{int} + \beta^{phy})$	22%	28%	36%	43%	21%	(2.90%)
Panel B: OLS R^2 values						
(1) i^{phy} on q^{tot}	0.219	0.227	0.259	0.284	0.065	(0.026)
(2) i^{int} on q^{tot}	0.061	0.170	0.306	0.458	0.397	(0.064)
(3) i^{tot} on q^{tot}	0.232	0.270	0.357	0.473	0.241	(0.016)
(4) CAPX/PPE on q^*	0.182	0.195	0.248	0.299	0.117	(0.022)
Panel C: ρ^2 values						
(1) i^{phy} on q^{tot}	0.261	0.364	0.486	0.612	0.351	(0.027)
(2) i^{int} on q^{tot}	0.271	0.311	0.377	0.411	0.140	(0.048)
(3) i^{tot} on q^{tot}	0.274	0.388	0.498	0.543	0.269	(0.020)
(4) CAPX/PPE on q^*	0.197	0.282	0.379	0.561	0.364	(0.023)
Panel D: τ^2 values						
(1) i^{phy} on q^{tot}	0.650	0.478	0.374	0.375	–0.275	(0.033)
(2) i^{int} on q^{tot}	0.196	0.365	0.561	0.792	0.596	(0.031)
(3) i^{tot} on q^{tot}	0.664	0.519	0.503	0.659	–0.005	(0.034)
(4) CAPX/PPE on q^*	0.682	0.514	0.483	0.439	–0.243	(0.062)
Panel E: Slopes on cash flow						
(5) i^{phy} on q^{tot}, c^{tot}	0.203	0.090	–0.009	–0.036	–0.239	(0.023)
(6) i^{int} on q^{tot}, c^{tot}	–0.018	0.018	0.060	0.110	0.128	(0.013)
(7) i^{tot} on q^{tot}, c^{tot}	0.227	0.148	0.100	0.129	–0.098	(0.024)
(8) CAPX/PPE on q^*, c^*	0.182	0.072	0.011	–0.003	–0.185	(0.026)
Number of observations	35,438	35,453	35,442	35,467		

Panel B). For example, the R^2 for the total-investment regression increases monotonically from 23% to 47%. Even when we use the literature's standard investment and q measures, the R^2 increase monotonically from 18% to 30%. This last result is surprising, because the standard q measure has more measurement error in firms with more intangibles: τ^2 is 44% in Quartile 4 versus 68% in Quartile 1. The patterns are similar when we compare manufacturing with high-intangible industries or the early with the late subperiod. The increases in R^2 across subsamples, tabulated in the last columns of Tables 4–6, are statistically significant for all four investment measures and in all three tables, with just two exceptions out of 12 (Table 5, Specification 1 and Table 6, Specification 1).

Second, ρ^2 values increase monotonically and roughly double when moving from the lowest to highest intangible quartile (Panel C). This result means that the true q 's explanatory power for investment is much stronger in firms with more intangible capital. This increase in ρ^2 is responsible for the large increases in R^2 across subsamples. Again, the patterns are similar across industries and years.

Third, cash flow slopes are significantly lower in firms, industries, and years with more intangible capital (Panel

E). The cash flow slopes even turn slightly negative in several high-intangible subsamples, even when we use the literature's standard measures. This result is robust across all four investment measures and across Tables 4–6, with one exception: Intangible investment has a larger cash flow slope in higher-intangible quartiles (Table 4). This exception could be an artifact of the measurement error bias we discuss in Section 4.2. Like us, Chen and Chen (2012) find a weaker investment-cash flow sensitivity in recent years. Our findings suggest this change over time could partially reflect the rise of intangible capital.

The rest of this subsection seeks to explain why the classic q theory works better in settings with more intangibles. Put differently, which of the theory's assumptions are violated more severely in firms using less intangibles? We start by exploring theoretically whether violations of our simple model's assumptions could explain the patterns in Table 4. We solve a more general model that relaxes our earlier assumptions about constant returns to scale, perfect competition, and quadratic adjustment costs. Details and numerical results are in the Online Appendix. We explain two predictions from the model next.

Table 5

Comparing industries.

This table shows results from industry subsamples. We use the Fama and French five-industry definition, excluding the industry “Other.” Remaining details are the same as in Table 4.

Specification	Manufacturing	Consumer	High-tech	Health	Quartile 4 – 1	
					Difference	Standard error
Intangible intensity	31%	48%	55%	62%		
Panel A: Slopes on q						
(1) ϵ^{phy} on q^{tot}	0.083	0.085	0.059	0.068	–0.015	(0.005)
(2) ϵ^{int} on q^{tot}	0.038	0.037	0.036	0.040	0.002	(0.003)
(3) ϵ^{tot} on q^{tot}	0.097	0.102	0.079	0.084	–0.013	(0.005)
(4) CAPX/PPE on q^*	0.041	0.042	0.033	0.038	–0.003	(0.003)
Panel B: OLS R^2 values						
(1) ϵ^{phy} on q^{tot}	0.194	0.239	0.307	0.244	0.050	(0.038)
(2) ϵ^{int} on q^{tot}	0.206	0.209	0.407	0.281	0.075	(0.031)
(3) ϵ^{tot} on q^{tot}	0.258	0.310	0.460	0.362	0.104	(0.024)
(4) CAPX/PPE on q^*	0.186	0.214	0.354	0.258	0.072	(0.031)
Panel C: ρ^2 values						
(1) ϵ^{phy} on q^{tot}	0.254	0.397	0.540	0.551	0.297	(0.036)
(2) ϵ^{int} on q^{tot}	0.321	0.234	0.474	0.376	0.055	(0.030)
(3) ϵ^{tot} on q^{tot}	0.294	0.386	0.572	0.521	0.227	(0.028)
(4) CAPX/PPE on q^*	0.206	0.290	0.549	0.545	0.339	(0.029)
Panel D: τ^2 values						
(1) ϵ^{phy} on q^{tot}	0.557	0.442	0.431	0.319	–0.238	(0.035)
(2) ϵ^{int} on q^{tot}	0.398	0.485	0.686	0.545	0.147	(0.041)
(3) ϵ^{tot} on q^{tot}	0.632	0.539	0.634	0.522	–0.110	(0.036)
(4) CAPX/PPE on q^*	0.655	0.539	0.511	0.365	–0.290	(0.048)
Panel E: Slopes on cash flow						
(5) ϵ^{phy} on q^{tot}, c^{tot}	0.171	0.029	–0.033	–0.059	–0.230	(0.032)
(6) ϵ^{int} on q^{tot}, c^{tot}	0.041	0.106	0.059	–0.019	–0.060	(0.017)
(7) ϵ^{tot} on q^{tot}, c^{tot}	0.265	0.190	0.090	0.010	–0.255	(0.034)
(8) CAPX/PPE on q^*, c^*	0.083	0.048	0.001	–0.003	–0.086	(0.018)
Number of observations	40,280	36,884	31,680	11,207		

We find that violating the assumption about quadratic adjustment costs is unlikely to generate the empirical patterns in Table 4. When we change the adjustment-cost function's exponent from 2 to 1.75 or 1.5, we find a negligible effect on predicted R^2 values, and we do not find a significant predicted investment–cash flow relation.

Differences in economies of scale or competition could theoretically explain some of the patterns in Table 4. Relative to the benchmark theory in Section 2, a theory with imperfect competition or decreasing returns to scale produces lower predicted R^2 values in regressions of investment on q , and it also generates a positive investment–cash flow relation, a prediction already known from Abel and Eberly (2011). If firms using more intangible capital are closer to the perfect-competition, constant-returns benchmark, this mechanism could explain why they exhibit lower cash flow slopes and higher R^2 and ρ^2 values.

Unfortunately, we find little empirical support for this mechanism. In the Online Appendix, we check whether firms with more intangibles are closer to the perfect-competition, constant-returns benchmark. First, we estimate production-function curvature using the methods of Cooper and Haltiwanger (2006) and Olley and Pakes (1996). Comparing the curvature estimates across subsamples, we find no statistically significant differences in economies of scale between the high- and low-intangible quartiles. Also, whereas the improvement in model fit

in Table 4 is monotonic across quartiles, the curvature estimates are strongly non-monotonic. Second, we compare three competition proxies across subsamples. We find mixed results when we use the Herfindahl Index to proxy for industry-level competition; different industry classifications deliver increasing, decreasing, or flat patterns across intangible-intensity subsamples. We also compare profitability across subsamples, because competition should reduce profitability. Again, different profitability measures produce opposing results. We also compare firm size across subsamples, as relatively small firms within an industry can face more competition. The relation between firm size and intangible usage is either statistically insignificant or strongly non-monotonic depending on the size proxy we use. To summarize, we do not find any robust empirical evidence that high-intangible firms face less diseconomies of scale or more competition.

One last possible explanation for the pattern in Table 4 is that high-intangible firms are less financially constrained, making the theory fit the data better. This explanation seems unlikely, because it is difficult to use intangible assets as collateral, which arguably makes high-intangible firms more financially constrained (Almeida and Campello, 2007; Falato, Kadyrzhanova, and Sim, 2013). Unfortunately, it is difficult to test this financing-constraints mechanism without a full structural estimation (Hennessy and Whited, 2007).

Table 6

Comparing time periods.

This table shows results from the early (1975–1995) and late (1996–2011) subsamples. The 1995 breakpoint produces subsamples of roughly equal size. Remaining details are the same as in Table 4.

Specification	Early	Late	Late - Early	
			Difference	Standard error
Intangible intensity	39%	47%		
Panel A: Slopes on q				
(1) ι^{phy} on q^{tot}	0.083	0.062	−0.021	(0.002)
(2) ι^{int} on q^{tot}	0.035	0.037	0.002	(0.002)
(3) ι^{tot} on q^{tot}	0.100	0.079	−0.021	(0.004)
(4) CAPX/PPE on q^*	0.043	0.033	−0.010	(0.001)
Panel B: OLS R^2 values				
(1) ι^{phy} on q^{tot}	0.205	0.208	0.003	(0.018)
(2) ι^{int} on q^{tot}	0.190	0.328	0.138	(0.016)
(3) ι^{tot} on q^{tot}	0.273	0.357	0.084	(0.013)
(4) CAPX/PPE on q^*	0.209	0.268	0.059	(0.017)
Panel C: ρ^2 values				
(1) ι^{phy} on q^{tot}	0.304	0.407	0.103	(0.016)
(2) ι^{int} on q^{tot}	0.259	0.497	0.238	(0.018)
(3) ι^{tot} on q^{tot}	0.336	0.511	0.175	(0.016)
(4) CAPX/PPE on q^*	0.262	0.479	0.217	(0.016)
Panel D: τ^2 values				
(1) ι^{phy} on q^{tot}	0.501	0.423	−0.078	(0.022)
(2) ι^{int} on q^{tot}	0.504	0.584	0.080	(0.030)
(3) ι^{tot} on q^{tot}	0.595	0.603	0.008	(0.022)
(4) CAPX/PPE on q^*	0.615	0.477	−0.138	(0.026)
Panel E: Slopes on cash flow				
(5) ι^{phy} on q^{tot}, c^{tot}	0.109	−0.033	−0.142	(0.017)
(6) ι^{int} on q^{tot}, c^{tot}	0.090	−0.033	−0.123	(0.013)
(7) ι^{tot} on q^{tot}, c^{tot}	0.256	0.038	−0.218	(0.020)
(8) CAPX/PPE on q^*, c^*	0.074	−0.008	−0.082	(0.009)
Number of observations	69,753	72,047		

5.2. Comparing adjustment costs across subsamples

Table 4 shows interesting patterns in q -slopes across subsamples. According to our theory, these q -slopes do not help us test our theory's predictions or assumptions. Instead, the q -slopes reflect adjustment-cost parameters.

Table 4 shows that firms using more intangibles have significantly smaller slopes of physical investment on q , and they have significantly larger slopes of intangible investment on q . The implications are that firms using more intangibles have physical capital that exhibits larger convex adjustment costs and that they have intangible capital that exhibits smaller convex adjustment costs.

This pattern in q -slopes points to differences in the nature of physical and intangible capital across firms, and it could also shed light on why some firms use more intangible capital. As we explain at the end of Section 2.1, if a firm's intangible capital is less costly than physical capital to adjust, then the firm is predicted to use relatively more intangible capital. As a result, firms using more intangible capital should have a higher intangible investment q -slope relative to the sum of slopes for physical and intangible investment. We show these slope ratios in Panel A of Table 4. The ratios increase monotonically across the quartiles, consistent with our theory. Our theory further predicts that the slope ratio equals firms' intangible intensity. The actual intensities, shown in the column labels, range from 8% to

76%, and the slope ratios range from only 22% to 43%. Our simple theory, therefore, explains part but not all of firms' different intangible-capital usage.

This exercise provides a useful consistency check on our theory. Some important caveats are in order, though. To link q -slopes to firms' optimal mix of capital types, our theory needs strong additional assumptions. The theory requires that physical and intangible capital are identical in all ways except for their quadratic adjustment-cost parameters. Outside our simple theory, alternate explanations could exist for the pattern we find in q -slopes across firms. We know from the investment- q literature that q -slopes need not reflect adjustment costs. For example, Abel and Eberly (2011) show that, even in a world with no adjustment costs, diseconomies of scale can make investment and Tobin's q positively related. Also, differences could exist between physical and intangible capital's purchase prices, depreciation rates, economies of scale, and adjustment-cost curvatures. These differences could affect both firms' optimal mix of capitals and their investment- q slopes. Because Table 4 does not control for these differences, we might just be picking up these omitted differences between physical and intangible capital.

To explore this potential bias further, we solve a more general model that allows physical and intangible capital to differ in ways not allowed in Section 2. We assume that physical and intangible capital share the same adjustment-cost parameters ($\gamma^{phy} = \gamma^{int}$), so we shut down the mechanism proposed above. We then ask whether other differences between physical and intangible capital could produce predicted q -slope patterns like the ones in Table 4. Details and numerical results are in the Online Appendix. First, we find that differences in purchase prices ($p_{it}^{phy} \neq p_{it}^{int}$) can explain why some firms use more intangible capital, but they do not explain why firms have different investment- q slopes. Second, we show that differences between the two capital types' economies of scale do not necessarily drive them to use more of one capital type and do not make their q -slopes differ significantly. These first two alternate explanations – differences in purchase prices or economies of scale – do not seem to work for the empirical patterns we find. Third, we show that if intangible capital depreciates faster than physical capital, then firms optimally use less intangibles and intangible investment has a slightly lower q -slope than physical investment, consistent with the patterns in Table 4. Finally, we relax the assumption that both capital types face quadratic adjustment costs. We show that if intangible capital faces less-convex adjustment costs than physical capital, then firms optimally use less intangible capital and intangible investment has a lower q -slope than physical investment, consistent again with the patterns in Table 4. We cannot rule out that these last two mechanisms, differences in depreciation rates or adjustment-cost convexities between the two capital types, are driving the Table 4 cross-sectional relation between q -slopes and capital mixes.

6. Macro results

The neoclassical theory of investment, including the theory in this paper, can easily be interpreted as a theory

of the macroeconomy, not a single firm. The macro literature has been interested in the investment- q relation going back to at least [Abel \(1980\)](#) and [Summers \(1981\)](#). We ask how this relation changes when we account for intangible capital.

Our macro sample contains 141 quarterly observations for the US economy from 1972Q2 to 2007Q2, the longest period for which all variables are available. Data on aggregate physical q and investment come from [Hall \(2001\)](#), who uses the flow of funds and aggregate stock and bond market data. The literature's standard q measure, again denoted q^* , is the ratio of the value of ownership claims on the firm, less the book value of inventories, to the reproduction cost of plant and equipment. The standard investment measure, again denoted i^* , equals private nonresidential fixed investment scaled by its corresponding stock, both of which are from the Bureau of Economic Analysis.

Data on the aggregate stock and flow of physical and intangible capital come from Carol Corrado and are discussed in [Corrado and Hulten \(2014\)](#). Earlier versions of these data are used by [Corrado, Hulten, and Sichel \(2009\)](#) and [Corrado and Hulten \(2010\)](#). Their measures of intangible capital include aggregate spending on business investment in computerized information (from NIPA), R&D (from the National Science Foundation and Census Bureau), and economic competencies, which include investments in brand names, employer-provided worker training, and other items. One advantage of these macro data relative to our firm-level data is that the macro data do not rely on an assumption about the fraction of SG&A representing an investment. As before, we measure the total capital stock as the sum of the physical and intangible capital stocks. We compute total q as the ratio of total ownership claims on firm value, less the book value of inventories, to the total capital stock. We define the investment rates i^{phy} , i^{int} , and i^{tot} as in our firm-level analysis. To mitigate problems from potentially differing data coverage, we use the [Corrado and Hulten \(2014\)](#) ratio of physical to total capital to adjust the [Hall \(2001\)](#) measures of physical q and investment.¹³

The correlation between standard and total q is extremely high, 0.997. The reason is that total q equals standard q times the ratio of physical to total capital, and this ratio has changed slowly and consistently over time ([Fig. 1](#)). Of more importance is the change from standard to total investment, which additionally requires multiplying i^* by the ratio of capital flows, which is much more volatile than the ratio of capital stocks. The correlation between total and standard investment is therefore much smaller, 0.43.

For comparison, we also use the [Philippon \(2009\)](#) aggregate bond q measure, which he obtains by applying a structural model to data on bond maturities and yields. Bond q is available at the macro level but not at the firm level. [Philippon \(2009\)](#) shows that bond q explains more of the aggregate variation in what we call physical

Table 7

Time series macro regressions.

Results are from 141 quarterly observations from aggregate US data, from 1972Q2 to 2007Q2. Each column uses a different investment measure. Standard q (q^*) equals the lagged aggregate stock and bond market value divided by the physical capital stock. [Hall \(2001\)](#) computes these measures from the flow of funds. Total q includes intangible capital by multiplying physical q by the ratio of physical to total capital. The ratio is from the [Corrado and Hulten \(2014\)](#) aggregate US data. Bond q is constructed by applying the structural model of [Philippon \(2009\)](#) to bond maturity and yield data. These data are from Philippon's website. Newey–West standard errors with autocorrelation up to 12 quarters are in parentheses. Standard errors for the ordinary least squares (OLS) R^2 values are computed via bootstrap.

	CAPX/ PPE (ι^*)	Investment scaled by total capital (K^{tot})		
		Physical (ι^{phy})	Intangible (ι^{int})	Total (ι^{tot})
Panel A: Regressions in levels				
Total q (q^{tot})		−0.001 (0.003)	0.019 (0.003)	0.017 (0.003)
Standard q (q^*)	0.002 (0.003)			
OLS R^2	0.035 (0.034)	0.014 (0.030)	0.570 (0.026)	0.610 (0.040)
Panel B: Regressions in levels with bond q				
Bond q	0.061 (0.009)	0.049 (0.011)	0.006 (0.039)	0.055 (0.032)
OLS R^2	0.462 (0.059)	0.347 (0.050)	0.001 (0.013)	0.139 (0.060)
Panel C: Regressions in four-quarter differences				
Total q (q^{tot})		0.007 (0.002)	0.004 (0.001)	0.01 (0.003)
Standard q (q^*)	0.007 (0.002)			
OLS R^2	0.124 (0.057)	0.106 (0.052)	0.096 (0.056)	0.121 (0.060)
Panel D: Regressions in four-quarter differences with bond q				
Bond q	0.056 (0.006)	0.043 (0.005)	0.017 (0.004)	0.060 (0.008)
OLS R^2	0.606 (0.053)	0.620 (0.059)	0.235 (0.074)	0.530 (0.070)

investment than standard q does. Bond q data are from Philippon's website.

[Fig. 2](#) plots the time series of investment and q . Panel A shows the standard q and investment measures, which omit intangible capital. Except in a few subperiods, q explains investment relatively poorly, as [Philippon \(2009\)](#) and others have shown. Panel B shows that the relation between total q and i^{phy} is still weak. Panel C shows a strong relation between total q and intangible investment, mainly because total q and intangible investment both trend up from 1982 to 2000. Panel D compares total investment and total q . Here the fit looks strongest of all.

To explore these patterns more carefully, [Table 7](#) shows results from time series regressions of investment on lagged q . Panel A shows regressions in levels, comparing our four investment measures. Consistent with [Fig. 2](#), the literature's standard measures and i^{phy} produce statistically insignificant q -slopes and R^2 values near zero. In stark contrast, intangible and total investment both have highly significant q -slopes, and they deliver R^2 values of 57% and 61%, respectively. These R^2 values are even higher than the 46% R^2 that [Philippon \(2009\)](#) obtains by regressing the standard investment measure on bond q (Panel B). Judging by R^2 , the classic q theory fits the data much better

¹³ To be precise, we use the [Hall \(2001\)](#) data on q^* and i^* and the [Corrado and Hulten \(2014\)](#) data on $A = K^{phy}/(K^{phy} + K^{int})$ and $B = I^{phy}/(I^{phy} + I^{int})$. We compute $q^{tot} = q^*A$, $i^{phy} = i^*A$, $i^{tot} = i^{phy}/B$, and $i^{int} = i^{tot} - i^{phy}$.

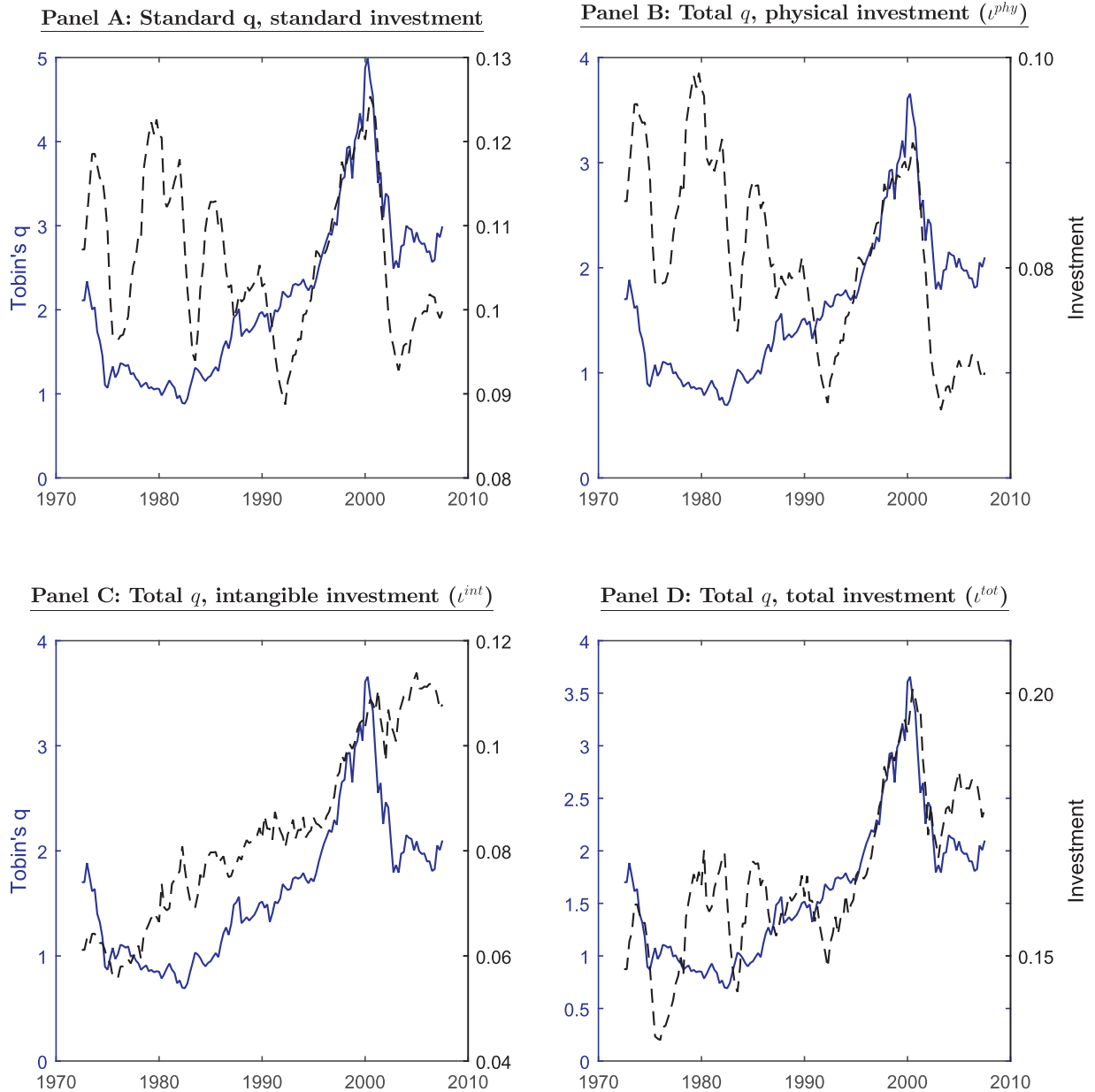


Fig. 2. Investment- q relation in macro data. This figure plots Tobin's q (solid lines) and the investment rates (dashed lines) over time for the aggregate US economy. Panel A uses data from Hall (2001) and shows standard measures that exclude intangible capital. Standard q (q^*) is aggregate market value scaled by the physical capital stock. Standard investment (l^*) equals physical investment scaled by the physical capital stock. Panels B–D also use data from Corrado and Hulten (2014). Total q is aggregate market value scaled by total capital, the sum of the physical and intangible capital stocks. Panel B shows l^{phy} , physical investment scaled by total capital. Panel C shows l^{int} , intangible investment scaled by total capital. Panel D shows $l^{tot} = l^{phy} + l^{int}$. For each graph, the left axis is the value of q and the right axis is the investment rate.

when we include intangible capital, because we are better able to explain the low-frequency trends in q and investment. Put differently, the literature's standard investment measure suffers from a low-frequency error—the omission of intangibles—that trends strongly with q over time.

How well can q explain higher-frequency variation in investment? Panels C and D answer this question by rerunning the previous regressions in four-quarter differences. As in our firm-level analysis, total q now explains physical and intangible investment roughly equally well, and it

explains total investment even better. As before, intangible investment has a lower q -slope than physical investment, indicating higher convex adjustment costs. Bond q , though, is much better than total q at explaining changes in investment. In differences, bond q also explains physical investment better than intangible investment. Philippon (2009) offers one potential explanation: Growth options affect stocks more than bonds, and growth options affect intangible investment more than physical investment. Put differently, physical and intangible capital can have

different values of marginal q . Bond q could be a better proxy for physical capital's marginal q , whereas the traditional q measures, which use stock prices, could be better proxies for intangible capital's marginal q . A second possible explanation is about sample selection. Firms with more intangible investment typically hold less debt, so they contribute less to the aggregate bond q measure.

To summarize, at the macro level, including intangibles makes q explain the level of investment much better, meaning the classic q theory fits the data much better than previously believed. When we try to explain changes in investment, the macro results look more like our firm-level results. Bond q is still better at explaining physical investment as well as changes in investment.

7. Robustness

This section describes our results' robustness across different subsamples, empirical measures, and estimators. We also explain why our main results are not mechanical.

7.1. Robustness of main results across subsamples

Tables 4–6 show that our main results are quite robust across subsamples. Compared with physical investment, intangible investment has a lower q -slope in all ten subsamples (Panel A). We always see larger q -slopes for physical investment in the specification with ι^{phy} compared with the specification with CAPX/PP&E (Panel A). Total q always explains total investment better than it explains either physical or intangible investment and better than standard q explains standard investment (Panel B). This result means that including intangibles produces a better proxy for investment opportunities even in subsamples with less intangible capital, such as the manufacturing industry. In the full sample (Table 3), total q explains intangible investment slightly better than physical investment. We see the reverse in four of ten subsamples, so we conclude that total q explains physical and intangible investment roughly equally well.

The improvement in model fit from including intangible capital is especially large in subsamples with more intangible capital, which is a useful consistency check. Consider the increase in R^2 when moving from the regression that ignores intangibles (Specification 4 in the tables) to the regression that uses ι^{tot} and q^{tot} (Specification 3). In Table 4, the increase in R^2 is 0.174 (58%) in the highest intangible quartile, but just 0.050 (27%) in the lowest quartile. This pattern is mainly driven by τ^2 , which increases by 0.284 (65%) in the top quartile but decreases by a statistically insignificant 0.018 (3%) in the lowest quartile. This result means that total q is a better proxy for true q , especially in firms with the most intangible capital. These same patterns are also present, but less dramatic, across industry and year subsamples.

7.2. What fraction of SG&A is an investment?

Arguably, the strongest assumptions in our intangible-capital measure are that $\lambda = 30\%$ of SG&A represents an investment and λ is constant across firms and time.

Table 8

Robustness: what fraction of SG&A expense is an investment?

Results are from regressions of three investment measures on lagged total q and firm fixed effects. Slopes on total q are from the cumulant estimator. Within-firm R^2 is from the ordinary least squares (OLS) estimator that also includes year fixed effects. The selling, general and administrative (SG&A) multiplier is the fraction of SG&A assumed to represent an investment. Our main analysis uses a 0.3 multiplier. For each multiplier value, we reestimate the intangible investment and capital stocks in the data. Because physical investment, total investment, and total q are scaled by total capital, their values also depend on the SG&A multiplier. Each regression uses 141,800 firm-year Compustat observations from 1975 to 2011. * indicates value used in main analysis

SG&A Multiplier	Investment scaled by total capital (K^{tot})					
	Physical (ι^{phy})		Intangible (ι^{int})		Total (ι^{tot})	
	q -slope	OLS R^2	q -slope	OLS R^2	q -slope	OLS R^2
0.0	0.060	0.223	0.021	0.147	0.064	0.277
0.1	0.064	0.217	0.025	0.256	0.074	0.307
0.2	0.067	0.213	0.032	0.276	0.081	0.320
0.3*	0.070	0.209	0.037	0.279	0.086	0.327
0.4	0.072	0.206	0.043	0.278	0.092	0.331
0.5	0.075	0.203	0.048	0.274	0.097	0.333
0.6	0.077	0.201	0.054	0.270	0.103	0.335
0.7	0.078	0.200	0.06	0.266	0.108	0.335
0.8	0.080	0.198	0.065	0.262	0.113	0.334
0.9	0.082	0.197	0.071	0.257	0.118	0.333
1.0	0.084	0.196	0.075	0.253	0.122	0.332

Table 8 shows that our main conclusions go though, at least qualitatively, when we use different values of λ ranging from zero to 100%. When λ is zero, firms' intangible capital comes exclusively from R&D. No matter what λ value we assume, we find larger q -slopes for physical investment, roughly equal R^2 for physical and intangible investment, and the highest R^2 for total investment. Intangible investment has its largest R^2 when $\lambda = 30\%$, meaning the data seem to prefer the λ value we use in our main analysis. The R^2 is considerably lower (15% versus 28%) if $\lambda = 0$, so the data do prefer counting at least part of SG&A as investment.

Instead of assuming $\lambda = 30\%$, we can let the data reveal λ 's value. The structural parameter λ affects both the investment and q measures. We estimate λ along with the q -slope and firm fixed effects by maximum likelihood, applied to the ι^{tot} regression. The estimated λ values are 0.38 in the consumer industry, 0.51 in the high-tech industry, and 0.24 in the health care industry, which are all in the neighborhood of our assumed 0.3 value. However, we do not push these λ estimates strongly, for three reasons. First, the investment- q relation is not the ideal setting for identifying λ . Second, the estimation imposes two very strong identifying assumptions: The linear investment- q model is true, and we measure all variables perfectly. Finally, the λ estimate in the manufacturing industry is constrained at 1.0, which is implausibly large and likely a symptom of the previous two issues. The main message from this subsection, though, is that our main conclusions hold regardless of the λ value we use.

7.3. Alternate measures of intangible capital

In addition to varying the SG&A multiplier λ , we try nine other variations on our intangible-capital measure.

Table 9

Robustness: alternate measures of intangible capital.

Results are from regressions of our three main investment measures on lagged total q and firm fixed effects. The column labels indicate the investment measure used. Slopes on q are from the cumulant estimator. We report the within-firm R^2 from the ordinary least squares (OLS) estimator that also includes year fixed effects. The first row reproduces results from Table 3 with our main intangible-capital measure. Rows 2–10 show results using alternate measures of intangible capital. Rows 2 and 3 use alternate values of $\delta_{SG\&A}$, the depreciation rate for organization capital. Rows 4–6 use alternate values of $\delta_{R\&D}$, the depreciation rate for knowledge capital. Row 7 excludes goodwill from balance sheet intangibles. Row 8 excludes all balance sheet intangibles. Row 9 assumes firms have no intangible capital before entering Compustat, which corresponds to setting $G_{i0} = 0$ in Eq. (11). Row 10 estimates firms' starting intangible capital using a perpetuity formula that assumes the firm has been alive forever before entering Compustat, as in Falato, Kadyrzhanova, and Sim (FKS 2013). The initial stock of knowledge capital (for example) is $G_{i0} = R\&D_{i1}/\delta_{R\&D}$, where $R\&D_{i1}$ is the research and development (R&D) amount in firm i 's first Compustat record. Row 11 uses our main intangible-capital measures but drops each firm's first five years of data. Row 12 uses our main intangible-capital measure but drops firm-year observations with missing R&D. Data are from Compustat from 1975 to 2011.

Specification	Investment scaled by total capital (K^{tot})						N
	Physical (ι^{phy})		Intangible (ι^{int})		Total (ι^{tot})		
	q -slope	OLS R^2	q -slope	OLS R^2	q -slope	OLS R^2	
1. Main results (Table 3)	0.070	0.209	0.037	0.279	0.086	0.327	141,800
2. $\delta_{SG\&A} = 10\%$	0.071	0.213	0.038	0.294	0.088	0.337	141,800
3. $\delta_{SG\&A} = 30\%$	0.069	0.207	0.037	0.270	0.086	0.322	141,800
4. $\delta_{R\&D} = 10\%$	0.073	0.209	0.038	0.285	0.089	0.329	141,800
5. $\delta_{R\&D} = 15\%$	0.072	0.209	0.038	0.282	0.088	0.328	141,800
6. $\delta_{R\&D} = 20\%$	0.070	0.208	0.038	0.279	0.087	0.327	141,800
7. Exclude Goodwill	0.070	0.209	0.037	0.282	0.086	0.329	141,800
8. Exclude balance sheet intangibles	0.063	0.199	0.035	0.248	0.079	0.306	141,800
9. Zero initial intangible capital	0.069	0.217	0.040	0.297	0.089	0.343	141,800
10. FKS initial multiplier	0.073	0.193	0.033	0.238	0.084	0.293	141,800
11. Drop first five years per firm	0.069	0.132	0.037	0.153	0.078	0.210	82,174
12. Exclude observations with missing R&D	0.062	0.267	0.036	0.359	0.081	0.411	75,426

We vary $\delta_{SG\&A}$, the depreciation rate for organization capital; we vary $\delta_{R\&D}$, the depreciation rate for knowledge capital; we exclude Goodwill from firms' intangible capital; we exclude all balance sheet intangibles, which brings us closer to existing measures from the literature; we set firms' starting intangible capital stock to zero; and we estimate firms' starting intangible capital stock using a perpetuity formula, like Falato, Kadyrzhanova, and Sim (2013). We also drop the first five years of data for each firm, which makes the choice of starting intangible capital stock less important. We also try dropping the 47% of firm-years with missing R&D from our regressions. Table 9 provides details about these variations and their results. Although magnitudes vary somewhat, our main results still hold in all these variations: Total q explains physical and intangible investment roughly equally well, total q explains total investment even better, and intangible investment always has a lower q -slope.

7.4. Alternate estimators

In addition to using the cumulant estimator to obtain unbiased q -slopes, we use the Biorn (2000) and Arellano and Bond (1991) instrumental variable (IV) estimators. Both estimators take first differences of the linear investment- q model and then use lagged regressors as instruments for the q proxy. Erickson and Whited (2012) show that these IV estimators are biased if measurement error is serially correlated, which is likely in our setting. This bias is probably most severe in the standard regressions that omit intangible capital, as omitting intangible capital is an important source of measurement error, and a firm's intangible capital stock is highly serially correlated. Because the cumulant estimators are robust to serially correlated measurement error, we prefer them over

the IV estimators. The IV estimators generate similar conclusions about adjustment costs. They produce lower q -slopes for ι^{int} than ι^{phy} and lower q -slopes for ι^* than ι^{phy} (Online Appendix).

7.5. A mechanical result?

Is it mechanical that total q explains total investment better than standard q explains standard investment? A potential concern is that moving from the latter regression to the former requires multiplying both sides of the regression by K^{phy}/K^{tot} . Multiplying both sides of a regression by the same variable can, but does not necessarily, increase the R^2 even if that variable is pure noise.

Our result is not mechanical or obvious, however. Multiplying both sides of the literature's standard regression by K^{phy}/K^{tot} produces the regression of ι^{phy} on q^{tot} , shown in Column 1 of Table 3. Contrary to the concern, that regression gets a slightly lower R^2 , τ^2 , and ρ^2 value than the standard regression (last column in Table 3). Moving to the regression of ι^{tot} on q^{tot} requires further multiplying ι^{phy} , but not q^{tot} , by the ratio of total to physical investment. This change would further reduce the R^2 if intangible investment were noise, but instead R^2 increases. Moreover, if our measure of intangible investment were just noise, we would not find that it is well explained by q and co-moves with physical investment. The Online Appendix presents a placebo simulation analysis showing that our main results would not obtain if our intangible capital measures were pure noise with similar statistical properties.

8. Conclusion

The neoclassical theory of investment has been applied almost exclusively to physical capital. We show that the

theory is also relevant for intangible capital, which increasingly dominates the US economy. In both our theory and firm-level data, physical and intangible investment co-move strongly, and they are explained roughly equally well by Tobin's q . Compared with physical capital, intangible capital's convex adjustment costs are roughly twice as large, meaning intangible capital responds more slowly to changes in investment opportunities. In macro data, Tobin's q explains the level of intangible investment many times better than physical investment. The neoclassical theory performs significantly better in firms, industries, and years with more intangible capital.

Tobin's q is "arguably the most common regressor in corporate finance" (Erickson and Whited, 2012, p. 1286). Guided by our theory, we provide a new Tobin's q measure that accounts for intangible capital, and we show that it is a superior proxy for both physical and intangible investment opportunities. This new Tobin's q measure offers a simple way to improve corporate finance regressions without additional econometrics. A benefit of the new measure is that it can be easily computed for the full Compustat sample. Data on our Tobin's q measure and firms' intangible capital can be downloaded from WRDS.

This paper revisits the basic facts about investment, Tobin's q , and cash flow while accounting for intangible capital. We believe this is an important step, because a vast investment literature in corporate finance and macroeconomics is built upon these facts. Important next steps include understanding how physical and intangible capital interact, how they face different prices for different firms in different periods, how they respond differently to growth options and financial constraints, and how they show up differently in firms' market values. Why the classic q -theory fits the data better in high-intangible settings is also an interesting open question. Finally, there is more work to do on measuring intangible capital.

Appendix A. Proofs

A.1. Proof of Prediction 1

Dropping firm subscripts, we can write the value function as

$$V_t = \max_{\{I_{t+s}^{phy}, I_{t+s}^{int}\}} \int_0^\infty E_t \left\{ K_{t+s}^{tot} \left[H(\varepsilon_{t+s}) - \frac{\gamma^{phy}}{2} \left(\frac{I_{t+s}^{phy}}{K_{t+s}^{tot}} \right)^2 - \frac{\gamma^{int}}{2} \left(\frac{I_{t+s}^{int}}{K_{t+s}^{tot}} \right)^2 - (p_{t+s}^{phy} + \zeta^{phy}) \frac{I_{t+s}^{phy}}{K_{t+s}^{tot}} - (p_{t+s}^{int} + \zeta^{int}) \frac{I_{t+s}^{int}}{K_{t+s}^{tot}} \right] \right\}. \quad (18)$$

Total capital follows

$$dK_t^{tot} = K_t^{tot} \left(\frac{I_t^{phy}}{K_t^{tot}} + \frac{I_t^{int}}{K_t^{tot}} - \delta \right) dt. \quad (19)$$

Following the same argument as in Appendix A of Abel and Eberly (1994), firm value must be proportional to total capital K^{tot} :

$$V(K^{phy}, K^{int}, \varepsilon, p^{phy}, p^{int}) = K^{tot} q^{tot}(\varepsilon, p^{phy}, p^{int}). \quad (20)$$

Differentiating this equation with respect to K^{phy} and K^{int} yields Eq. (5).

A.2. Proof of Prediction 2

Following a similar proof as in Abel and Eberly (1994), one can derive the Bellman equation and take first-order conditions with respect to each investment rate to obtain

$$q_t^{tot} = \frac{\partial}{\partial I_t^m} c^m(I_t^m, K_t^{tot}, p_t^m) = p_t^m + \zeta^m + \gamma^m \frac{I_t^m}{K_t^{tot}}, \quad m = phy, int. \quad (21)$$

Rearranging yields Eqs. (6) and (7).

A.3. Proof of Prediction 4

Multiplying both sides of Eq. (6) by $K_{it}^{tot}/K_{it}^{phy}$ yields

$$l_{it}^* = \frac{I_{it}^{phy}}{K_{it}^{phy}} = \frac{1}{\gamma^{phy}} \left(q_{it}^* - \frac{K_{it}^{tot}}{K_{it}^{phy}} (\zeta_i^{phy} + p_{it}^{phy}) \right). \quad (22)$$

Now consider a regression of l_{it}^* on q_{it}^* and firm and time FEs. The residual in that regression, ε_{it}^* , equals the residual from a regression of $-\frac{1}{\gamma^{phy}} \frac{K_{it}^{tot}}{K_{it}^{phy}} (\zeta_i^{phy} + p_{it}^{phy})$ on firm and time FEs. This residual is nonzero and, hence, the regression's R^2 is less than 100%, because the ratio $K_{it}^{tot}/K_{it}^{phy}$ cannot be fully explained by firm and time fixed effects. To see this last claim, define $\omega_{it} = K_{it}^{tot}/K_{it}^{phy}$. By Ito's lemma, ω evolves according to

$$\frac{d\omega_{it}}{\omega_{it}} = [l_{it}^{phy} (1 - \omega_{it}) + l_{it}^{int}] dt. \quad (23)$$

The evolution of ω_{it} cannot be fully explained by firm and time FEs, because it depends on the investment rates l_{it}^{phy} and l_{it}^{int} , which depend on q_{it}^{tot} and, hence, ε_{it} , which cannot be fully explained by the FEs. Furthermore, the error term ε_{it}^* is negatively correlated to the regressor $q_{it}^* = q_{it}^{tot} K_{it}^{tot}/K_{it}^{phy}$, because $K_{it}^{tot}/K_{it}^{phy}$ multiplies both terms, albeit with a negative sign in ε_{it}^* . Because the error term is negatively related to the regressor, the regression produces downward-biased estimates of $1/\gamma^{phy}$.

A.4. Proof of last prediction

Set $d\omega_{it} = 0$ in Eq. (23) and solve for the equilibrium value, $\bar{\omega}$:

$$\begin{aligned} \bar{\omega} &= \frac{l_{it}^{int} + l_{it}^{phy}}{l_{it}^{phy}} = \frac{\frac{1}{\gamma^{int}} (q_{it}^{tot} - \zeta - p_{it}) + \frac{1}{\gamma^{phy}} (q_{it}^{tot} - \zeta - p_{it})}{\frac{1}{\gamma^{phy}} (q_{it}^{tot} - \zeta - p_{it})} \\ &= \frac{\gamma^{phy} + \gamma^{int}}{\gamma^{int}}. \end{aligned} \quad (24)$$

The last prediction follows, because $K^{int}/K^{tot} = 1 - 1/\bar{\omega}$.

Appendix B. Measuring intangible capital

B.1. Measuring SG&A

We measure SG&A as Compustat variable $xsga$ minus xrd minus $rdip$. We add the following screen: When xrd exceeds $xsga$ but is less than $cogs$, or when $xsga$ is missing,

we measure SG&A as $xsga$ with no further adjustments or zero if $xsga$ is missing.

The logic behind this formula is as follows. According to the Compustat manual, $xsga$ includes R&D expense unless the company allocates R&D expense to cost of goods sold (COGS). For example, $xsga$ often equals the sum of Selling, General and Administrative and Research and Development on the Statement of Operations from firms' 10-K filings. To isolate (non-R&D) SG&A, we must subtract R&D from $xsga$ when Compustat adds R&D to $xsga$. There is a catch: When a firm externally purchases R&D on products not yet being sold, this R&D is expensed as In-Process R&D and does not appear on the balance sheet. Compustat adds to $xsga$ only the part of R&D not representing acquired In-Process R&D, so our formula subtracts $rdip$ (In-Process R&D Expense), which Compustat codes as negative. We find that Compustat almost always adds R&D to $xsga$, which motivates our formula above. Standard & Poor's explained in private communication that, "in most cases, when there is a separately reported xrd , this is included in $xsga$." As a further check, we compare the Compustat records and 10-K filings for a random sample of one hundred firm-year observations with non-missing xrd . We find that Compustat includes R&D in $xsga$ in 90 out of one hundred cases, partially includes it in $xsga$ in one case, and includes it in COGS in seven cases. Two cases remain unclear even after asking the Compustat support team. The screen above lets us identify obvious cases in which xrd is part of COGS. This screen catches six of the seven cases in which xrd is part of COGS. Unfortunately, identifying the remaining cases is impossible without reading SEC filings. We thank the Compustat support team from Standard & Poors for their help in this exercise.

We set $xsga$, xrd , and $rdip$ to zero when missing. For R&D and SG&A, we make exceptions in years when the firm's assets are also missing. For these years, we interpolate these two variables using their nearest non-missing values. We use these interpolated values to compute capital stocks but not regressions' dependent variables.

B.2. Measuring firms' initial capital stock

This section explains how we estimate the stock of knowledge and organization capital in firm i 's first non-missing Compustat record. We describe the steps for estimating the initial knowledge-capital stock. The method for organization capital is similar. Broadly, we estimate firm i 's R&D spending in each year of life between the firm's founding and its first non-missing Compustat record, denoted year one. Our main assumption is that the firm's pre-IPO R&D grows at the average rate across pre-IPO Compustat records. We then apply the perpetual inventory method to these estimated R&D values to obtain the initial stock of knowledge capital at the end of year zero. The seven steps are as follows.

1. Define age since IPO as number of years elapsed since a firm's IPO. Using the full Compustat database, compute the average log change in R&D in each yearly category of age since IPO. Apply these age-specific growth rates to fill in missing R&D observations before 1977.
2. Using the full Compustat database, isolate records for firms' IPO years and the previous two years. (Not all firms have pre-IPO data in Compustat.) Compute the average log change in R&D within this pre-IPO subsample, which equals 0.348. (The corresponding pre-IPO average log change in SG&A equals 0.333.)
3. If firm i 's IPO year is in Compustat, go to Step 5. Otherwise go to the next step.
4. This step applies almost exclusively to firms with IPOs before 1950. Estimate firm i 's R&D spending in each year between the firm's IPO year and first Compustat year, assuming the firm's R&D grows at the average age-specific rates estimated in Step 1.
5. Obtain data on firm i 's founding year from Jay Ritter's website. For firms with missing founding year, estimate the founding year as the minimum of (a) the year of the firm's first Compustat record and (b) firm's IPO year minus eight, which is the median age between founding and IPO for IPOs from 1980 to 2012 (from Jay Ritter's website).
6. Estimate the firm i 's R&D spending in each year between the firm's founding year and IPO year assuming the firm's R&D grows at the estimated pre-IPO average rate from Step 2.
7. Assume the firm is founded with no capital. Apply the perpetual inventory method in Eq. (11) to the estimated R&D spending from the previous steps to obtain G_{i0} , the stock of knowledge capital at the beginning of the firm's first Compustat record.

We only use estimated R&D and SG&A values to compute firms' initial stocks of intangible capital. We never use estimated R&D in a regression's dependent variable.

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